Shock Syndromes II: Hypovolemic, Critical Bleeding, and Obstructive

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Learning Objectives

1. Identify critical determinants affecting oxygen delivery and the physiologic response to hypovolemic and obstructive shock.
2. Evaluate resuscitation strategies and end points in the management of hypovolemic, hemorrhagic, and obstructive shock.
3. Devise a treatment strategy for pharmacotherapy adjuncts in the management of bleeding and acute coagulopathy when treating patients with hemorrhagic shock.
4. Develop a treatment pathway for the care of patients receiving anticoagulants and antiplatelet agents for a life-threatening hemorrhage or critical bleeding that incorporates current evidence and guideline recommendations.
5. Apply risk stratification to guide the effective and safe use of thrombolytic agents in the management of acute pulmonary embolism.

Abbreviations in This Chapter

4F-PCC	4-factor prothrombin complex concentrate
ABC	Assessment of Blood Consumption
ACC	American College of Cardiology
aPCC	Activated prothrombin complex concentrate
aPTT	Activated partial thromboplastin time
CDT	Catheter-directed thrombolysis
CO	Cardiac output
DOAC	Direct oral anticoagulant
Do_2	Oxygen delivery
FAST	Focused Assessment with Sonography in Trauma
Fio_2	Fraction of inspired oxygen
GCS	Glasgow Coma Scale (score)
LV	Left ventricular
MTP	Massive transfusion protocol
PCC	Prothrombin complex concentrate
PE	Pulmonary embolism
PH	Pulmonary hypertension
PRBC	Packed red blood cell
PT	Prothrombin time
rFVIIa	Recombinant activated factor VII
RV	Right ventricular
TBSA	Total body surface area
TEG	Thromboelastogram
TTE	Transthoracic echocardiogram

Self-Assessment Questions

Answers and explanations to these questions may be found at the end of this chapter.

1. A 41-year-old man presents to the emergency department (ED) after a motorcycle accident. While the patient is being evaluated, it is apparent that he has broken ribs, a broken pelvis, and bilateral broken femurs. He is confused, and his vital signs are as follows: blood pressure 82/48 mm Hg, heart rate 125 beats/minute, respiratory rate 34 breaths/minute, and temperature 95°F (35°C). On further assessment, laboratory tests show that the patient has lost around 35% of his total blood volume. Which most accurately reflects this patient's class of hemorrhage from trauma?

   A. I.
   B. II.
   C. III.
   D. IV.

2. A 62-year-old man with a history of alcoholic cirrhosis and portal hypertension presents to the ED with weakness and hematemesis. Pertinent vital signs on admission are as follows: blood pressure 76/42 mm Hg, heart rate 134 beats/minute with a rhythm of sinus tachycardia, and respiratory rate 24 breaths/minute. Frank red blood is noted on nasogastric lavage. On physical examination, the patient is confused, with cold and clammy extremities. Pertinent laboratory values are as follows: hemoglobin (Hgb) 5.5 g/dL, blood urea nitrogen (BUN) 56 mg/dL, serum creatinine (SCr) 1.8 mg/dL, international normalized ratio (INR) 2.6, total bilirubin 7.5 mg/dL, central venous oxygen saturation ($Scvo_2$) 58%, and lactate 2.7 mmol/L. In this patient, which variable would best explain reduced oxygen delivery (Do_2) and which associated clinical or biochemical sign would best indicate a shock syndrome?

   A. Elevated INR and hematemesis.
   B. Sinus tachycardia and elevated serum creatinine.
   C. Low hemoglobin and cold and clammy extremities.
   D. Low $Scvo_2$ and hyperlactatemia.

3. A 44-year-old man (weight 82 kg) presents to the ED after a motor vehicle collision at highway speeds secondary to his windshield being shattered by a rock thrown from an overpass. On primary survey, his King airway device is replaced for an endotracheal tube; his respiratory rate is 37 breaths/minute, systolic blood pressure (SBP) is 77 mm Hg, heart rate is 146 beats/minute, and Glasgow Coma Scale score is 3. In addition, he is noted to have a positive seatbelt sign. The Focused Assessment with Sonography in Trauma (FAST) examination is positive. He is taken to the operating room for an emergency exploratory laparotomy. Which best represents the most appropriate initial transfusion, resuscitation, and hemostasis strategy?
   A. Give 2 L of warmed lactated Ringer solution, followed by a 2-unit transfusion of erythrocytes.
   B. Administer 30 mL/kg of warmed lactated Ringer solution and a tranexamic acid 1-g bolus, followed by 1 g infused over 8 hours.
   C. Obtain and send an INR, activated partial thromboplastin time (aPTT), fibrinogen, and complete blood cell count to guide initial resuscitation.
   D. Initiate massive transfusion, giving erythrocytes and plasma in a 1:1 ratio plus a tranexamic acid 1-g bolus and 1 g infused over 8 hours.

4. H.S. is a 76-year-old man (weight 142 kg) with a history of several pulmonary embolisms (PEs) and associated chronic thromboembolic pulmonary hypertension (CTEPH) on warfarin who is admitted to the ED with weakness and hematemesis. Pertinent vital signs on admission are as follows: blood pressure 82/46 mm Hg, heart rate 121 beats/minute with a rhythm of sinus tachycardia, and respiratory rate 22 breaths/minute. Frank red blood is noted on nasogastric lavage. On physical examination, the patient is confused, with clammy extremities. Pertinent laboratory values are as follows: Hgb 5.2 g/dL, BUN 52 mg/dL, SCr 2 mg/dL, INR 9.2, and platelet count (Plt) 120,000/mm^3. Which is most effective to immediately reverse his coagulopathy secondary to warfarin?
   A. Phytonadione 10 mg orally once.
   B. 4-factor prothrombin complex concentrate (4F-PCC) 50 units/kg intravenously infused over 30 minutes.
   C. Fresh frozen plasma 15-mL/kg infusion once.
   D. Phytonadione 10-mg intravenous push once.

5. A 66-year-old man with a medical history of non–small cell lung cancer presents to the ED with new-onset shortness of breath. A chest computed tomography (CT) scan reveals a PE at the bifurcation of the right and left pulmonary arteries. The patient is initiated on parenteral anticoagulation and transferred to the medical intensive care unit (ICU). On ICU admission, he develops pulseless electrical activity. He is intubated and mechanically ventilated, with recovery of spontaneous circulation after one round of chest compressions and epinephrine 1 mg. Which is the next best step to evaluate and/or treat this patient's PE?
   A. Administer alteplase 100 mg infused over 2 hours.
   B. Check a troponin T concentration.
   C. Check a brain natriuretic peptide concentration.
   D. Obtain a transthoracic echocardiogram (TTE).

Questions 6–8 pertain to the following case.

R.M is a 78-year-old man with a history of alcoholic cirrhosis and portal hypertension who is in an urban, academic ICU after treatment of a variceal hemorrhage (now stable) and respiratory failure. Five days after admission, he develops hypoxemia (partial pressure oxygen saturation [Pao_2] 78%), requiring an increased fraction of inspired oxygen (Fio_2) on the ventilator. A chest computed tomography angiography (CTA) reveals several filling defects at the bifurcation of the main pulmonary artery, suggesting a "saddle" PE. His heart rate is 142 beats/minute with a rhythm of sinus tachycardia. His blood pressure is 97/62 mm Hg and current weight is 87 kg. R.M. has a TTE that reveals right ventricular (RV) hypokinesis and tricuspid regurgitation. His cardiac troponin I (0.6 ng/mL) and troponin T (0.2 ng/mL) are positive.

6. Which would best categorize R.M.'s PE?
   A. High risk.
   B. Intermediate-high risk.
   C. Intermediate-low risk.
   D. Low risk.

7. After initiating therapeutic heparin and monitoring, R.M.'s blood pressure is now maintained at 90/50 mm Hg, but his oxygenation continues to worsen, now requiring 80% F_{IO_2}. The medical team has decided that the patient's potential benefit from thrombolytic therapy outweighs the potential risks. Which is the best approach for this patient's thrombolytic administration?
   A. Alteplase 100 mg infused over 2 hours.
   B. Catheter-directed thrombolysis (CDT).
   C. Alteplase 50 mg infused over 2 hours.
   D. Alteplase 0.5 mg/hour by a pulmonary artery catheter.

8. As you prepare to begin administering your strategy chosen previously, R.M. goes into a cardiac arrest with pulseless electrical activity. Which is the best intervention for managing R.M.'s PE while still in the ICU?
   A. Give alteplase 50 mg intravenous push once.
   B. Proceed to surgical embolectomy.
   C. Give alteplase 100 mg over 2 hours.
   D. Abort cardiopulmonary resuscitation.

9. K.A. is a 50-year-old man admitted to the neonatal ICU because of a head trauma causing a subarachnoid hemorrhage. On ICU day 3, his blood pressure is 70/50 mm Hg, and he is mechanically ventilated. K.A. was resuscitated with a normal saline bolus, and norepinephrine was initiated. A TTE reveals RV dilation and tricuspid regurgitation, and a CT scan reveals a filling defect within the pulmonary arteries, dilatation of the right ventricle of the heart, and enlargement of the pulmonary artery. K.A.'s condition continues to worsen, and vasopressin is added for refractory shock. Which is the best approach for treatment of K.A.'s shock?
   A. Place an inferior vena cava filter.
   B. Administer alteplase 100 mg infused over 2 hours.
   C. Administer alteplase 50 mg infused over 2 hours.
   D. Consult for possible catheter-directed thrombolysis.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
   a. Task A: 1, 2, 4, 6
2. Domain 2: Therapeutics and Patient Management
   a. Task A: 1, 2, 4–6, 8
   b. Task B: 1, 2, 6, 10, 13, 15, 16
   c. Task C: 1–4
3. Domain 3: Professional Practice
   a. Task A: 5
   b. Task B: 1, 7

   Systems and patient care problems:
   - Hypovolemic shock
   - Acute traumatic coagulopathy
   - Burn resuscitation
   - Antithrombotic reversal
   - Obstructive shock
   - Social determinants of health
   - Disaster recovery

I. HYPOVOLEMIC SHOCK

A. Etiology and Epidemiology

1. Hypovolemic shock is a result of reduced intravascular volume (i.e., reduced preload), which, in turn, reduces stroke volume and cardiac output (CO), leading to inadequate tissue perfusion. Hypovolemic shock accounts for around 16% of all cases of shock requiring vasoactive medications.
2. In the United States, the most common form of shock secondary to trauma is hypovolemic shock, and exsanguination is estimated to be the direct cause of around 50,000 deaths from trauma in the United States (Table 1).
   a. Responsible for up to 30%–40% of trauma-related mortality
   b. Leading cause of death in patients younger than 45; thus, burden to society is even larger
3. Although commonly associated with trauma, hypovolemic shock can also occur in other clinical scenarios (e.g., heat stroke, acute gastrointestinal [GI] bleeding, surgical, obstetrics, pharmacologic toxicity, burns, pancreatitis, loss of extracellular fluid).
4. A study examined the impact of race and insurance status on trauma mortality rates among patients 18–64 years of age with an Injury Severity Score of 9 or higher caused by blunt or penetrating trauma, using data from the National Trauma Data Bank from 2001 to 2005. Specifically, the study aimed to determine whether race and insurance status independently predicted outcome disparities after trauma. The crude mortality rates were 5.7%, 8.2%, and 9.1% for the White, African American, and Hispanic groups, respectively ($p<0.005$). Crude mortality by insurance status was statistically lower for patients with insurance (4.4%) than for patients without insurance (8.6%) ($p<0.005$). The study also found that African American and Hispanic patients with insurance had higher mortality rates than White patients with insurance, and this effect worsened for patients without insurance across groups.

Table 1. Estimated Hemorrhage-Related Deaths/yr and Years of Life Lost in the United States and Worldwide, According to the Cause of Hemorrhage

Cause of Hemorrhage	Death from Hemorrhage (%)[a]	U.S. Cases		Global Cases	
		No. of Deaths/yr	Years of Life Lost	No. of Deaths/yr	Years of Life Lost
Trauma	30	49,440	1,931,786	1,481,700	74,568,000
Abdominal aortic aneurysm	100	9988	65,273	191,700	2,881,760
Peptic ulcer disease	60	1860	38,597	141,000	3,903,600
Maternal disorder	23	138	7572	69,690	4,298,240

[a]Estimates of deaths from hemorrhage as a percentage of all deaths from the given diagnosis.

Modified from: Cannon JW. Hemorrhagic shock. N Engl J Med 2018;378:370-9.

B. Pathophysiology

1. Hypovolemic shock can be categorized as hemorrhagic and nonhemorrhagic:
   a. Whole blood loss (hemorrhagic): Whole blood loss from an open wound or into a body compartment (e.g., limb, retroperitoneal space)
   b. Plasma loss (nonhemorrhagic): Loss of extracellular fluid (e.g., burns, pancreatitis, third spacing, peritonitis, vomiting, diarrhea)
2. The estimated blood volume for a patient weighing 70 kg is 5 L (75 mL/kg for men and 65 mL/kg for women). Hemorrhagic shock occurs when intravascular volume loss impairs Do_2, generally greater than 30% of total blood volume loss.

3. Clinical features of hypovolemic shock include hypotension, tachycardia, diaphoresis, altered mentation, and decreased urinary output. If hypovolemic shock is secondary to blood loss from trauma, physiologic variables may be used to estimate the extent of blood loss (Table 2).

Table 2. Classification of Trauma Hemorrhage[a]

	Class I	Class II (mild)	Class III (moderate)	Class IV (severe)
Blood volume loss (%)	< 15%	15%–30%	30%–40%	> 40%
Heart rate	↔	↔/↑	↑	↑/↑↑
Blood pressure	↔	↔	↔/↓	↓
Pulse pressure	↔	↓	↓	↓
Respiratory rate	↔	↔	↔/↑	↑
Urine output	↔	↔	↓	↓↓
Glasgow Coma Scale score	↔	↔	↓	↓
Base deficit	0 to -2 mEq/L	-2 to -6 mEq/L	-6 to -10 mEq/L	-10 mEq/L or less

[a]Classification system is only intended as a guide to initial therapy because the physiologic response to hemorrhage represents a continuum. Confounding factors that influence the physiologic response to hemorrhage include patient age, severity of injury, time from injury, prehospital interventions, and medications for chronic conditions. Therefore, it is not intended to wait for a patient to fit each precise physiologic classification before initiating volume resuscitation.

Information from: 10th Edition of the Advanced Trauma Life Support® (ATLS®) Student Course Manual. American College of Surgeons, 2018.

4. Physiologic response
   a. Compensatory responses try to restore the volume deficit.
   b. Neural response is immediate, occurring within minutes.
      i. Sympathetic response: Activation of the low-pressure receptors within the right and left atria and high-pressure receptors within the aortic arch and carotid sinus lead to increased secretion of epinephrine and norepinephrine, resulting in increased heart rate, myocardial contractility, and arteriolar/venous tone. Blood flow is preserved to critical organs.
      ii. Parasympathetic response: Reduced vagal tone leads to increased heart rate. Often, tachycardia is the earliest sign of circulatory shock from acute blood loss.
   c. Intrinsic response compensates for acute blood loss within hours.
      i. Reduced capillary pressure leads to fluid redistribution from the interstitial space to the vascular compartment as albumin shifts into the plasma.
      ii. The transcapillary refill can recruit up to 1 L into the intravascular compartment.
   d. Humoral response is delayed, developing over hours to several days. After decreased renal perfusion, secretion of antidiuretic hormone, aldosterone, and renin increases sodium and volume retention to restore the interstitial deficit from the transcapillary refill.

C. Acute Traumatic Coagulopathy (ATC):
   1. There is no consensus regarding values defining traumatic coagulopathy, and various definitions exist. However, it is commonly defined as an increase in standard plasma-based coagulation tests above normal limits not associated with antiplatelet or anticoagulation therapy (aPTT, prothrombin time [PT], and INR). These include PT greater than 18 seconds, INR greater than 1.5, aPTT greater than 60 seconds, or any of these values at a threshold of 1.5 times the laboratory reference value. The prevalence of prolonged PT is higher, but prolongation of the PTT is more specific. In a multicenter, observational study, an INR-based definition of acute coagulopathy (INR greater than 1.5) was significantly associated with death (odds ratio [OR] 1.88; $p<0.001$), venous thromboembolism (OR 1.73; $p<0.001$), and multiorgan failure (OR 1.38; $p=0.02$).

2. Present in 25%–35% of injured civilian trauma patients on arrival at the ED and associated with a 40%–70% mortality rate, which likely plays a significant role in preventable deaths caused by hemorrhage
3. Pathophysiology of ATC is multifactorial:
    a. Inadequate tissue perfusion as the result of hypovolemia
    b. Subsequent cell hypoxia, anaerobic respiration, and metabolic acidosis
    c. Thrombin–thrombomodulin-complex generation caused by the tissue injury and the activation of anticoagulant and fibrinolytic pathways
    d. Hyperfibrinolysis plays a central role in the initial coagulopathy.
    e. Severe systemic hypothermia is associated with decreases in the enzymatic activity of clotting factors.
    f. Coagulopathy, acidosis, and hypothermia—known as the "lethal triad"—exacerbate one another, rapidly leading to death if not reversed.
    g. The DISPARITY study investigated whether sex differences existed in Surviving Sepsis Campaign resuscitation bundle completion, completion of individual bundle elements, or sepsis mortality. The study aimed to identify potential disparities in ED care for patients with sepsis on the basis of sex. The study reported that women with coagulopathy (INR greater than 1.5) may have received less aggressive care than men. The study showed a specific association of coagulopathy with bundle noncompletion in women (but not men) with an adjusted odds ratio of 0.49 (95% CI, 0.25–0.94). However, the analysis of sex-specific factors in survival showed a trend toward an association between bundle completion and survival in women but not men.
4. Characterization of ATC
    a. Standard coagulation laboratory or viscoelastic tests are recommended as monitoring techniques to characterize ATC (European Trauma Guidelines, grade 1C recommendation).
    b. Viscoelastic tests play an emerging role in screening for coagulopathy in hemorrhagic shock.
        i. Whole blood coagulation tests that represent global clot function, including clot initiation, formation, stabilization, and fibrinolysis, correlate well with standard plasma coagulation studies and may be available as point-of-care tests with a faster turnaround time.
        ii. Available tests include thromboelastogram (TEG) and rotational thromboelastometry (ROTEM), which are more commonly used in the United States and Europe, respectively.
        iii. TEG outputs include a graphic representation of clot kinetics (Figure 1 and Figure 2).

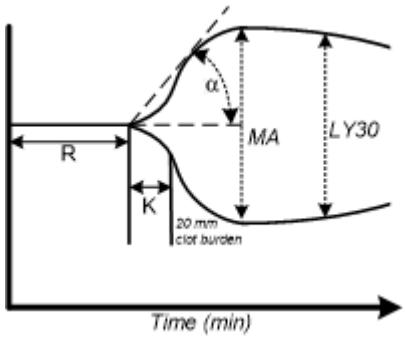


Figure 1. Features of a thromboelastogram.

α = α- angle; K = kinetics; LY30 = lysis at 30 minutes; MA = maximum amplitude; R = reaction (time).

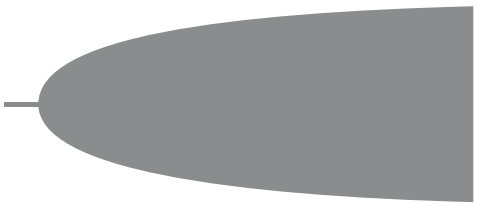
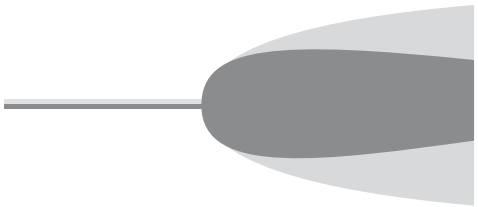
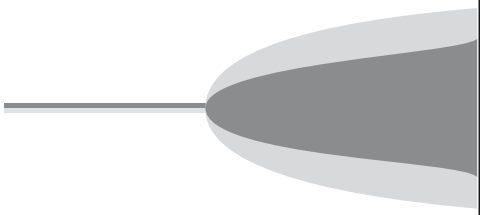
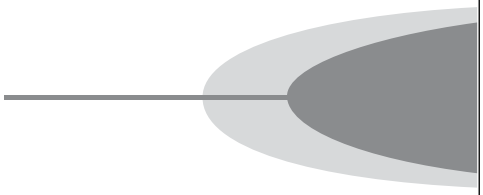
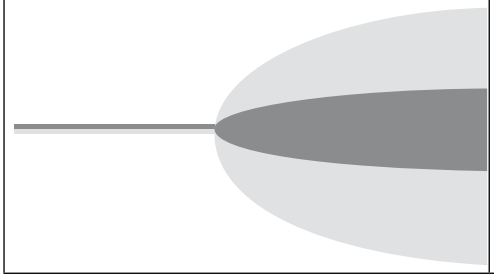
TEG Tracing	Interpretation	Treatment
	Hypercoagulable	Enoxaparin or heparin prophylaxis when possible
	Hyperfibrinolysis	Tranexamic acid
	Fibrinogen deficiency	Cryoprecipitate -or- Concentrated fibrinogen
	Factor deficiency	Plasma -or- prothrombin complex concentrate
	Platelet dysfunction	Platelet transfusions -or- desmopressin

Figure 2. Examples of TEG tracings (dark) relative to the control (light) with corresponding interpretation and treatments.

Table 3. TEG Output Parameters and Normal Values

TEG Parameters	Description	Normal Range[a] (kaolin-activated)	Treatment
Reaction (R) time	Time until evidence of a first clot is detected. Time of latency from start to initial fibrin formation because of effects of factor VIIa and tissue factor (time until fibrin clot formation)	3–9 min	Fresh frozen plasma if elevated
Kinetics (K)	Time from the end of R (factor initiation and start of clot formation) until the clot reaches 20 mm amplitude of clot strength (speed of clot formation), which represents platelet activity, factor levels, and fibrinogen	1–3 min	Cryoprecipitate if elevated
α-Angle	The speed at which fibrin buildup and cross-linking take place (clot strengthening) and hence assesses the rate of clot formation	55–78 degrees	Cryoprecipitate if decreased
Maximum amplitude (MA)	Represents the strongest point of fibrin clot, predominantly platelets but also fibrinogen (80% of the MA is derived from platelet function, and the remaining 20% is derived from fibrin)	51–69 mm	Platelets if decreased
Estimated percent lysis or lysis at 30 min (LY30)	Represents the percent decrease in amplitude 30 min after MA and measures the degree of fibrinolysis	0%–8%	Aminocaproic acid if increased

[a]Standard TEG is kaolin-activated, and values differ if native blood is used. TEG using native whole blood is slower (R-time 4–8 min, α-angle 47–74 degrees, MA 55–73 mm, K 1–4 min, and LY30 typically not reported or interpreted).

TEG = thromboelastography.

Information from: Gonzalez E, Pieracci FM, Moore EE, et al. Coagulation abnormalities in the trauma patient: the role of point-of-care thromboelastography. Semin Thromb Hemost 2010;36:723-37.

D. Coagulopathy of Chronic Liver Disease
   1. Patients with chronic liver disease often have episodes of clinically meaningful bleeding events confounded by decreased concentrations of most endogenous procoagulant factors, except for factor VIII and von Willebrand factor.
   2. Historically, the basic laboratory tests of coagulation (e.g., PT, aPTT) have been used to determine the risk of bleeding, despite their poor correlation with onset and duration of bleeding after liver biopsy or predicting the occurrence of GI hemorrhage.
   3. This is likely explained by the parallel reduction in endogenous anticoagulants (e.g., antithrombin, protein C) in chronic liver disease, leading to a "balanced state of coagulation."
   4. Randomized controlled trials evaluating recombinant activated factor VII (rFVIIa) to reverse the coagulopathy of chronic liver disease showed dramatically improved PT, but rFVIIa failed to alter bleeding complications during liver transplantation or control bleeding in variceal hemorrhage.

5. The American Association for the Study of Liver Diseases (AASLD) 2017 updated guideline for managing portal hypertensive bleeding in cirrhosis recommends against the use of fresh frozen plasma or rFVIIa to correct an INR because INR is not a reliable indicator of coagulation in cirrhosis, and evidence is lacking that this improves outcomes. In addition, the American Gastroenterological Association (AGA) 2021 guidelines on coagulation in cirrhosis state that the balanced nature of alterations in hemostasis associated with end-stage liver disease is complex and that an elevated INR is not necessarily predictive of bleeding outcomes in patients with cirrhosis. For patients with stable cirrhosis (with known baseline abnormal coagulation parameters) undergoing common GI procedures, the AGA suggests against the use of extensive preprocedural testing (including repeated measurements of PT/INR or Plt) and against the routine use of blood products (e.g., fresh frozen plasma and Plt) for bleeding prophylaxis.
6. Global clotting tests (e.g., the thrombin generation test or TEG) have the potential to characterize coagulopathy in chronic liver disease but may not be routinely available, and optimal transfusion thresholds remain undefined.

E. Resuscitation
   1. Rapid identification and correction of the source of bleeding is the definitive treatment (e.g., surgical exploration, angiographic embolization, stabilization of the pelvic ring, damage control surgery).
   2. Fluids
      a. Indications: Diminished mental status or absent radial pulse (SBP less than 90 mm Hg)
      b. Benefit: Fluids restore intravascular volume, reverse tissue hypoperfusion, and correct oxygen debt.
      c. Risk: Fluids do not increase oxygen-carrying capacity, can precipitate dilutional coagulopathy, and lead to tissue edema, including the development of acute respiratory distress syndrome.
      d. Crystalloids: Balanced crystalloids (e.g., lactated Ringer solution) and isotonic saline are both recommended as resuscitation fluids. Isotonic saline is most commonly used, but it can cause hyperchloremic metabolic acidosis and resultant acute kidney injury. Balanced crystalloids are associated with less hyperchloremia in trauma patients, but it is unknown whether this improves morbidity and mortality. Hypertonic saline resuscitation has failed to improve outcomes.
      e. Colloids: Confer no incremental benefit over crystalloids, were associated with increased mortality in a subgroup analysis of patients with severe brain injury in the SAFE trial, likely secondary to hypo-osmolarity. Synthetic colloids (e.g., hydroxyethyl starches) contribute to coagulopathy and the risk of acute kidney injury and should therefore be avoided.
      f. Recommendations: Isotonic crystalloids are indicated in hemorrhagic shock. In trauma, restricted volume replacement, usually less than 1.5 L of a balanced crystalloid, should be used initially, and hypotonic solutions (e.g., lactated Ringer with a sodium content of 131 mmol/L) should be avoided in patients with head trauma to minimize fluid shifts into the cerebral tissue. The most recent European guidelines on managing major bleeding and coagulopathy after trauma recommend isotonic balanced crystalloids over saline.
   3. Packed red blood cells (PRBCs) and blood products
      a. Indicated when the estimated blood loss is greater than 30% of the total blood volume. Each PRBC unit is expected to increase circulating Hgb by around 1 g/dL.
      b. Amount of blood products to transfuse is based on clinical examination, given that the initial hemoglobin or hematocrit reading may not reflect blood loss because of compensatory mechanisms.
      c. Although there are no randomized controlled trials evaluating transfusion thresholds for trauma, the European guidelines on managing major bleeding and coagulopathy after trauma recommend maintaining a hemoglobin of 7–9 g/dL after initial resuscitation.

d. In acute upper GI bleeding, a restrictive transfusion threshold (Hgb less than 7 g/dL) compared with a liberal transfusion threshold (Hgb less than 9 g/dL) was associated with a higher 6-week survival rate (95% vs. 91%, hazard ratio [HR] 0.55 [95% confidence interval [CI], 0.33–0.92; p=0.02]) and lower rates of further bleeding (10% vs. 16%, p=0.01) and adverse effects (40% vs. 48%, p=0.02).
   i. Notable limitations of the study include its single-center design and exclusion of patients with cardiovascular disease.
   ii. Despite these limitations, a transfusion threshold of 7 g/dL was endorsed by the 2017 AASLD Practice Guidance for Portal Hypertension Bleeding in Cirrhosis, with recommendations to maintain hemoglobin at 7–9 g/dL.

e. A randomized controlled trial of patients undergoing cardiac surgery compared a restrictive transfusion threshold (less than 7.5 g/dL) with a liberal transfusion threshold (less than 9 g/dL).
   i. The groups did not differ with respect to the composite primary end point of serious infection or ischemic event (35.1% vs. 33.0%; OR 1.11; 95% CI, 0.91–1.34; p=0.30).
   ii. However, more deaths occurred in the restrictive transfusion group (4.2% vs. 2.6%; HR 1.64; 95% CI, 1.00–2.67; p=0.045).
   iii. Therefore, a restrictive transfusion strategy after cardiac surgery cannot be recommended.

f. A study examined transfusion practices in hospitalized general medicine patients and tested for differences in the Hgb concentration at which patients have transfusions and the total number of RBC units received by patients' race. The study aimed to determine any disparities in transfusion practices according to race. In this study, African Americans had lower transfusion rates than Whites (25% vs. 30%, respectively; p<0.01). For Hgb concentrations below a nadir Hgb of 9 g/dL, African Americans had significantly lower transfusion rates than Whites. The transfusion rate for Hgb concentrations of 8.0–8.9 g/dL was 1% for African American patients versus 7% for White patient (p<0.01). The transfusion rate for Hgb concentrations of 7.0–7.9 g/dL was 15% and 28% for African Americans and Whites, respectively (p<0.01). The transfusion rate for Hgb concentrations below 7 g/dL was 80% for African Americans versus 86% for Whites (p<0.01). African American patients also received fewer units of RBCs overall, with a beta coefficient of -0.17 (p<0.01), and at lower Hgb concentrations, with a beta coefficient of 0.14 (p<0.01), than White patients. These observed differences in the receipt of transfusion by race met the definition of a health care disparity.

4. Vasopressors
   a. Attractive adjuncts in hemorrhagic shock to minimize the amount of fluid required to reverse tissue hypoperfusion, but can increase cardiac afterload and are independently associated with increased mortality in trauma
   b. Similar to other shock states, a relative arginine vasopressin deficiency develops in hemorrhagic shock that is associated with catecholamine resistance, vasoplegia, and increased venous capacitance. Supplemental vasopressin may be an adjunct to blood product resuscitation in hemorrhagic shock.
      i. The AVERT-Shock trial was a single-center, randomized controlled trial investigating the impact of low dose vasopressin (up to 0.04 units/min), compared to placebo, in adult trauma patients who required at least 6 units of any blood product within 12 hours of injury. They found that patients who received vasopressin required significantly less cumulative volume of all blood products by around 1 L, and had a decreased incidence of deep vein thrombosis. No difference was noted in other complications, such as acute respiratory distress syndrome, or outcomes, including mortality.
      ii. Although this study challenges the traditional practice of avoiding vasopressors in a bleeding trauma patient, confirmatory research is needed to determine if vasopressin significantly improves morbidity and mortality.
   c. In life-threatening hypotension, vasopressors may be recommended only after hypovolemia has been corrected or when cardiac arrest is imminent.

5. End points of resuscitation
    a. General recommendations include permissive hypotension, defined as SBP 80–90 mm Hg for most patients, and urinary output greater than 30 mL/hour. Up to 85% of patients may be under-resuscitated using SBP and urinary output as end points.
    b. In patients with traumatic brain injury (TBI), permissive hypotension should be avoided. The 2016 Brain Trauma Foundation guidelines recommend maintaining SBP at 110 mm Hg or greater for patients 15–49 years of age or older than 70 and at 100 mm Hg or greater in patients 50–69 years of age.

Patient Case

1. A 67-year-old man is accidentally shot in the buttocks while deer hunting with his friends. He is brought to the ED immediately. While in transfer, the patient receives 500 mL of lactated Ringer solution. His vital signs on admission to the ED are as follows: blood pressure 104/58 mm Hg, heart rate 108 beats/minute, respiratory rate 22 breaths/minute, and temperature 95°F (35°C). On examination, he has 8/10 pain and appears anxious. Which is best for resuscitation strategies?
    A. Administer lactated Ringer solution at 1000 mL/hour to maintain a urinary output greater than 30 mL/hour and an SBP greater than 100 mm Hg.
    B. Transfuse 2 units of PRBCs and administer a 1-L bolus of lactated Ringer solution to maintain a urinary output greater than 1 mL/kg/hour.
    C. Transfuse 2 units of PRBCs, 2 units of fresh frozen plasma, and a 1-L bolus of lactated Ringer solution to normal mentation.
    D. Administer lactated Ringer solution at 1000 mL/hour to maintain a urinary output greater than 30 mL/hour, an SBP greater than 80–90 mm Hg, and normal mentation.

F. Burn Resuscitation
    1. Acute thermal injury triggers an inflammatory state that ultimately leads to third spacing of intravascular fluid.
    2. Fluid resuscitation is initiated to maintain perfusion to tissue beds and end-organ function. A concomitant concern is to avoid over-administration of fluids, leading to abdominal or extremity compartment syndrome, acute respiratory distress syndrome, and further third spacing.
    3. The 2008 American Burn Association (ABA) practice guidelines for burn shock resuscitation promote fluid resuscitation to target a urine output of 0.5–1 mL/kg/hour in adults, with the initial rate determined by the total body surface area (TBSA) burned and body weight. Given a lack of strong evidence, the ABA recommends against a preload-driven resuscitation target from an invasive monitor unless special circumstances exist (e.g., older patients or inadequate response to treatment).
    4. Many formulas exist to estimate crystalloid need, but the most common strategy is administering crystalloid fluid using lactated Ringer solution by the Parkland formula: 4 mL x kg x % TBSA, with half administered over the first 8 hours and the remaining half over the next 16 hours.
    5. In practice, significantly more fluid is often given to burn patients than is predicted by traditional formulas.
        a. While maintaining a urinary output more than 0.5 mL/kg/hour, the average patient with a severe burn receives around 6 mL/kg/% body surface area within the first 24 hours, leading to the potential for "fluid creep" and associated complications of over-resuscitation.
        b. Therefore, some formulas now advocate less initial fluid, such as the "modified" Brooke formula (2 mL/kg/% body surface area).

c. The U.S. Army Institute of Surgical Research (USAISR) rule of 10 (% TBSA burn x 10 = initial rate) is a simplified equation that determines the initial starting rate for fluid resuscitation, and subsequent rate changes are guided by maintaining a urinary output greater than 0.5 mL/kg/hour.

6. Burn resuscitation formulas serve as an initial starting dose, and ongoing resuscitation should be individualized.
7. Difficult to resuscitate: When a patient's fluid requirement exceeds what is expected according to the traditional formulas, albumin may be added. When added, the current fluid rate is maintained, but albumin replaces 33%–66% of the total rate. High dose ascorbic acid (e.g., 66 mg/kg/hr) has been shown to decrease total fluid requirements, but guidelines state more evidence is needed before this therapy can be recommended routinely.

Patient Case

2. A 29-year-old man (height 72 inches, weight 85 kg) is admitted to the burn unit with a 40% TBSA burn to the lower extremities and buttocks after a fall into a molten slag at work in a steel mill. He received 500 mL of normal saline during transfer to the hospital. On his presentation to the burn unit, the surgical resident asks for your help in calculating the patient's fluid resuscitation needs. Which is best for resuscitation?
   A. 15 L of lactated Ringer solution over 24 hours; initiate at 1000 mL/hour for the first 12 hours, followed by 250 mL/hour, titrating to goal urinary output of 1 mL/kg/hour.
   B. 13 L of lactated Ringer solution over 24 hours; initiate at 813 mL/hour for the first 8 hours, followed by 406 mL/hour, titrating to goal urinary output of 0.5 mL/kg/hour.
   C. 12 L of lactated Ringer solution over 12 hours; initiate at 1000 mL/hour, titrating to goal urinary output of 0.5 mL/kg/hour.
   D. 24 L of lactated Ringer solution over the first 24 hours; initiate at 1000 mL/hour, titrating to goal urinary output of 0.5 mL/kg/hour.

G. Blood Product Resuscitation
   1. Lethal triad: Hypothermia, acidosis, and coagulopathy
      a. Hypothermia, severe acidemia (pH less than 7.20), and hypocalcemia inhibit the procoagulant enzyme function.
      b. Therefore, treatment of the acutely bleeding patient should prioritize warming the patient to a temperature greater than 93.2°F (34°C), correcting acidosis to a pH greater than 7.20, and administering calcium to an ionized calcium greater than 4.4 mg/dL (1.1 mmol/L).
   2. Initial bleeding and coagulopathy should be managed with blood component therapy (e.g., plasma, platelets, cryoprecipitate) to build whole blood, termed *damage control resuscitation*.
      a. Standard massive transfusion protocols include at least a 1:1:1 strategy of PRBCs/plasma/platelets.
      b. The multicenter, randomized PROPPR trial compared a balanced 1:1:1 damage control resuscitation transfusion strategy of plasma/platelets/PRBCs with a ratio of 1:1:2 in adult trauma patients.
         i. 1:1:1 group received 6 units of plasma, one dose of platelets (6 units of platelets), and 6 units of PRBCs. Transfusion order was platelets first, followed by alternating plasma and PRBC units.
         ii. 1:1:2 group received an initial container (and all subsequent odd-numbered containers) with 3 units of plasma, no doses of platelets, and 6 units of PRBCs. The second (and all subsequent even-numbered containers) had 3 units of plasma, one dose of platelets (6 units of platelets), and 6 units of PRBCs. Transfusion order was platelets first, followed by alternating 2 units of PRBCs and 1 unit of plasma.
         iii. 24-hour mortality (12.7% in 1:1:1 group vs. 17.0% in 1:1:2 group; adjusted RR 0.75; 95% CI, 0.52–1.08; p=0.12) and 30-day mortality (22.4% in 1:1:1 group vs. 26.1% in 1:1:2 group; adjusted RR 0.86; 95% CI, 0.65–1.12; p=0.26) did not differ significantly.

iv. However, in the 1:1:1 arm, hemostasis was achieved in more patients (86.1% vs. 78.1%, p=0.006), possibly highlighting the importance of early platelet transfusion, and fewer experienced death caused by exsanguination (9.2% vs. 14.6%; p=0.03).

3. Application of damage-control resuscitation in the prehospital setting has been investigated to intervene as close to the time of injury as possible to prevent the development of coagulopathy, irreversible shock, and the ensuing inflammatory response.
   a. The Prehospital Air Medical Plasma (PAMPer) trial was designed to determine the efficacy and safety of prehospital administration of 2 units of plasma resuscitation with or without red blood cells compared with standard of care (crystalloids with or without red blood cells) in severely injured patients at risk of hemorrhagic shock.
      i. Patients were eligible if they had at least one episode of hypotension (SBP less than 90 mm Hg), tachycardia (heart rate greater than 108 beats/minute), or severe hypotension (SBP less than 70 mm Hg) in the prehospital setting.
      ii. 501 patients were enrolled, with a median prehospital transport time of 40–42 minutes.
      iii. After arrival at the trauma center, the PT ratio was lower in the plasma group (1.2 vs. 1.3; p<0.001). Mortality rates at 24 hours (13.9% vs. 22.1%; p=0.02) and 30 days (23.2% vs. 33%; p=0.03) were lower in the plasma group, with early separation in the Kaplan-Meier survival curve at 3 hours. There was no difference in inflammation-mediated complications, including multiorgan failure, acute respiratory distress syndrome, or nosocomial infections, nor did transfusion-related reactions differ, suggesting prehospital plasma is safe.
   b. In contrast, the benefits of prehospital plasma may not be as apparent in an urban setting with short prehospital transport times. In a single-center study at Denver Health Medical Center, prehospital plasma failed to improve mortality at 28 days (15% plasma vs. 10% control; p=0.37) and was terminated early for futility after 144 enrollments.
   c. Although logistical challenges are related to the storage and shelf life of plasma in a variety of preparations, trauma centers may consider implementing prehospital plasma for patients at risk of hemorrhagic shock, particularly in rural settings with longer transport times.

H. Massive Transfusion Protocol (MTP)
1. Massive transfusion: defined as a transfusion of 10 units or more of RBCs in 24 hours, 3 units of RBCs over 1 hour, or any four blood components in 30 minutes
2. Focused Assessment with Sonography in Trauma (FAST) instrument can be used to evaluate trauma patients with suspected abdominal and thoracoabdominal injuries and detect hemorrhages in the peritoneal, pleural, and pericardial cavities.
3. Clinical scores have been developed to assist in early identification of patients who would benefit from activation of the MTP. The Assessment of Blood Consumption (ABC) Score was utilized in the PROPPR trial for study enrollment. Two or more of the following criteria are associated with high sensitivity and specificity for requiring an MTP: penetrating mechanism of injury, ED SBP 90 mm Hg or less, ED heart rate 120 beats/minute or greater, or a positive FAST examination.
4. Current recommendations for the initial management of an expected massive hemorrhage include PRBCs and plasma in a ratio of at least 2:1 as needed, or fibrinogen concentrate and PRBCs (European Trauma Guidelines 2019; grade 1C).
5. Further resuscitation, including platelet transfusions, should be guided in a goal-directed fashion, using either standard laboratory coagulation assays or viscoelastic tests (grade 1B). An example of goal-directed resuscitation can be found in Table 4.
6. In a prospective study of 111 patients, TEG-guided resuscitation was superior to standard laboratory coagulation-guided resuscitation with improved 6-hour mortality (7.1% vs. 21.8%; p=0.032) and total mortality (19.6% vs. 36.4%; p=0.049) and required less plasma and platelets in the first 2 hours of resuscitation. Notable limitations include single-center study design and TEG being available at point of care.

Table 4. Example of Goal-Directed Resuscitation

Conventional Coagulation Assay (CCA)	Thromboelastography (TEG)
INR ≥ 1.5: 2 units of FFP Fibrinogen < 150 mg/dL: 10 pack of cryoprecipitate Plt < 100,000/mm^3: 1 unit of apheresis platelets D-dimer > 0.5 mcg/mL: 1 g of tranexamic acid	Initial activated clotting time ≥ 140 s: 2 units of FFP, 10 pack of cryoprecipitate, 1 unit of apheresis platelets Activated clotting time > 111–139 s[a]: 2 units of FFP Alpha angle < 63: 10 pack of cryoprecipitate MA < 55 mm: 1 unit of apheresis platelets LY30[b] > 3%: 1 g of tranexamic acid

[a]Equivalent to R-time > 10

[b]LY30 is a reflection of fibrinolysis, similar to estimated percent lysis.

FFP = fresh frozen plasma; LY30 = percent decrease in amplitude after 30 minutes; MA = maximum amplitude; Plt = platelets.

Information from: Gonzalez E, Moore EE, Moore H, et al. Goal directed resuscitation of trauma induced coagulopathy: a pragmatic randomized clinical trial comparing a viscoelastic assay to conventional coagulation assay. Ann Surg 2016;263:1051-9.

7. Some metabolic complications are associated with massive transfusions that are notable for pharmacotherapy considerations.
    a. Potassium abnormalities may include hypokalemia or hyperkalemia.
        i. Hyperkalemia:
            (a) ATPase pumps become deactivated in stored blood, leading to elevations in potassium concentrations from 7 mEq/L to 77 mEq/L.
            (b) After transfusion, the ATPase pump is restored, and potassium is shifted intracellularly.
            (c) Although this generally leads to minimal sequelae, hyperkalemia may develop in those with renal insufficiency or severe tissue injury and transfusion rates. Rapid transfusion through a central venous catheter has been associated with hyperkalemia-induced cardiac arrest.
            (d) For patients at risk, the recommended management is to slow the infusion rate to less than 100–150 mL/minute.
        ii. Hypokalemia: Secondary to restoration of the ATPase pump on transfusion, shifting potassium intracellularly, metabolic alkalosis from citrate administration, release of hormones (e.g., aldosterone, antidiuretic hormone), and co-infusion of potassium-poor solutions (e.g., normal saline)
        iii. Potassium should be closely monitored and managed during massive transfusion.
    b. Complications secondary to citrate:
        i. Each unit of stored blood is anticoagulated with around 3 g of citrate. The metabolic capacity of the liver for citrate metabolism is around 3 g every 5 minutes, which is diminished with underlying liver dysfunction or in shock states.
        ii. Accumulation of citrate after several transfusions may bind endogenous calcium, inducing severe hypocalcemia, prolonged QT intervals, circulatory depression, hypotension, tremors, and pulseless electrical activity.
        iii. Therefore, it is critically important to monitor ionized calcium concentrations and to administer calcium chloride or calcium gluconate to maintain normal concentrations.
        iv. Evidence is currently lacking regarding optimal calcium dosing in a massive transfusion. Calcium chloride 1–3 g is often the preferred dose and preparation in the setting of MTP due to the higher potency of elemental calcium relative to calcium gluconate. The total elemental calcium concentration in calcium chloride 10% and calcium gluconate 10% is 270 mg/10 mL and 90 mg/10 mL, respectively.
        v. Magnesium may also be bound by citrate, which can further increase the effects of hypocalcemia, necessitating appropriate monitoring and management.

vi. Although citrate is metabolized to bicarbonate, which may induce metabolic alkalosis, with prolonged storage, PRBCs have a pH below 7.0, contributing to metabolic acidosis. Metabolic acidosis is more common in patients requiring a massive transfusion, given hypoperfusion and anaerobic metabolism in a shock state.

I. Antifibrinolytics
   1. Tranexamic acid
      a. An antifibrinolytic agent that binds plasminogen to prevent the dissolution of a fibrin clot. Tranexamic acid is a competitive inhibitor of plasmin and plasminogen.
      b. CRASH-2 trial:
         i. Evaluated the effects of tranexamic acid in adult trauma patients with, or at risk of, significant bleeding who were within 8 hours of injury compared with placebo (0.9% saline). Tranexamic acid (loading dose of 1 g over 10 minutes, followed by a 1-g infusion over 8 hours) reduced 28-day mortality for all-cause trauma compared with placebo (relative risk [RR] 0.91; 95% CI, 0.85–0.97; p=0.0035).
         ii. The number needed to treat is 67, and no difference between groups was noted in the risk of vascular occlusive events.
         iii. In a separately published exploratory analysis, the greatest benefit for tranexamic acid occurred with administration within 3 hours of injury (RR 0.68 1 hour or less from injury; 95% CI, 0.57–0.82; p<0.0001; RR 0.79 1–3 hours from injury; 95% CI, 0.64–0.97; p=0.03), with an increased risk of death from bleeding after 3 hours from injury (RR 1.44; 95% CI, 1.12–1.84; p=0.004).
         iv. Major limitations of this trial were that few patients were enrolled from regions with modern trauma systems (e.g., blood product availability, massive transfusion protocol), and the baseline mortality was higher than expected in North America or Europe.
      c. CRASH-3 trial:
         i. Evaluated the effects of tranexamic acid (same dosing regimen as used in CRASH-2) in patients with TBI. Patients were enrolled within less than 3 hours of injury with either Glasgow Coma Scale (GCS) score less than 13 or intracranial hemorrhage on CT scan and no major extracranial bleeding.
         ii. There was no difference in head injury–related death within 28 days with tranexamic acid (18.5%) compared with placebo (19.8%) (RR 0.94; 95% CI, 0.86–1.02).
         iii. Tranexamic acid reduced death in patients with mild to moderate TBI, GCS 9–15 (RR 0.78; 95% CI, 0.64–0.95), but not in patients with a severe head injury, GCS 3–8 (RR 0.99; 95% CI, 0.91–1.07) who already had extensive intracranial hemorrhage before treatment.
         iv. Patients in CRASH-3 were sicker than those in CRASH-2, with head injury–related death 12% in CRASH-2 versus 19% in CRASH-3.
      d. In the MATTERs (Military Application of Tranexamic Acid in Trauma Emergency Resuscitation) study, tranexamic acid was associated with a 6.5% absolute reduction in in-hospital mortality for all patients. A 13.7% absolute reduction occurred in those requiring massive transfusion, suggesting a greater benefit in those with a higher severity of injury.
      e. In the MATTERs II trial, adding fibrinogen (cryoprecipitate) to tranexamic acid further improved in-hospital mortality compared with tranexamic acid alone (11.6% vs. 18.2%; p=0.03).
      f. In trauma, tranexamic acid has not been shown to decrease the need for transfusion.
      g. Recommendations related to tranexamic acid use in trauma:
         i. When given, should be administered as early as possible to a trauma patient with bleeding or within 3 hours of injury to a patient at risk of significant hemorrhage. Administered as a loading dose of 1 g infused over 10 minutes, followed by an intravenous infusion of 1 g over 8 hours

ii. Some hospital systems have begun giving the tranexamic acid loading dose prehospital to prevent full activation of fibrinolysis, with improved outcomes when the loading dose was given within 1 hour of injury. This strategy requires validation in a clinical trial.
iii. Additional safety concerns have been identified related to tranexamic acid in trauma patients.
(a) Although not shown in initial trials, tranexamic acid was subsequently associated with thrombotic events, including PE and venous thromboembolism (MATTERs trial).
(b) In addition, tranexamic acid may be associated with fibrinolysis shutdown, a state of suppressed fibrinolysis associated with high mortality from multiorgan failure.
iv. Because of criticisms of previous trials and potential safety concerns, some experts advocate the use of tranexamic acid only in select patients having a diagnosis of hyperfibrinolysis, but no study on selective use of tranexamic acid has shown improved outcomes.
v. In the United States, the Eastern Association for the Surgery of Trauma (EAST) guidelines for damage control resuscitation conditionally recommended tranexamic acid as a hemostatic adjunct in severely injured trauma patients.

h. Tranexamic acid in postpartum hemorrhage:
i. In the WOMAN trial, a tranexamic acid 1-g bolus (which may be repeated at 30 minutes) did not affect all-cause mortality but did reduce mortality from postpartum hemorrhage if administered within 3 hours compared with placebo (1.2% vs. 1.7%; RR 0.69; 95% CI, 0.52–0.91; p=0.008).
ii. The World Health Organization strongly recommends tranexamic acid within 3 hours for postpartum hemorrhage.

i. The importance of early tranexamic acid was evaluated in an integrated analysis of both the WOMAN trial and the CRASH-2 trial. For every 15-minute delay to tranexamic acid, the overall benefit was decreased by around 10%, highlighting the importance of early administration.

j. In cardiac surgery, perioperative tranexamic acid reduced blood product transfusions and rates of major hemorrhage without a higher risk of death from thrombotic complications, but with an increased risk of seizures (0.7% vs. 0.1%; p<0.01).
i. Of note, the tranexamic acid doses of 50–100 mg/kg used in this study are much higher than those used in trauma.
ii. The increased risk of seizures may be unique to cardiac surgery because the risk of stroke and seizures has been described after cardiac surgery, independent of tranexamic acid use.

2. Aminocaproic acid
a. A lysine analog that is 10-fold weaker than tranexamic acid
b. Recommended dose is a 150-mg/kg bolus, followed by a 15-mg/kg/hour infusion.
c. Compared with tranexamic acid, aminocaproic acid may cause more adverse drug reactions, including hypotension and bradycardia, with rapid administration and rhabdomyolysis.
d. Lower quality of evidence exists for aminocaproic acid in the treatment of the acutely bleeding patient.

J. Recombinant Activated Factor VIIa (rFVIIa)
1. Activates hemostasis through the extrinsic pathway of the coagulation cascade by interacting with tissue factor to activate factor X to factor Xa and factor IX to factor IXa. Adequate concentrations of platelets and fibrinogen are needed to support activity.
2. Case series and case reports suggest that rFVIIa improves clinical outcomes.
3. In trauma, a randomized controlled trial (CONTROL trial) was terminated early because of the futility for the primary end point of mortality, though investigators did find a reduction in PRBC transfusion with rFVIIa (200 mcg/kg followed by 100 mcg/kg at 1 hour and 3 hours) in patients with blunt trauma.

4. There is no consensus on rFVIIa use for acute bleeding. Institutional protocols vary with respect to appropriate indication, timing, dosing, and readministration of rFVIIa.
5. Database reviews document an increased incidence of venous and arterial thromboembolic events with off-label rFVIIa use for non-hemophilia indications.
6. In trauma, rFVIIa should only be considered if major bleeding and traumatic coagulopathy persist despite all other attempts to control bleeding and optimize the physiologic environment, including correction in acidosis, hypothermia, and hypocalcemia.

K. Prothrombin Complex Concentrates (PCCs)
   1. A combination of concentrated clotting factors II, VII, IX, and X and proteins C and S
      a. Four different forms of PCC are available in the United States. Factor content of the PCCs varies, particularly for factor VII (Table 5).
      b. FEIBA (factor eight inhibitor bypassing activity) is the only PCC composed of activated clotting factors, making it an activated prothrombin complex concentrate (aPCC). Although theoretically more efficacious, this may also lead to increased thrombotic events.
   2. Kcentra, a nonactivated 4F-PCC, has U.S. Food and Drug Administration (FDA)-approved labeling for use in the reversal of warfarin-related acute bleeding disorders (further discussed under Reversal of Oral Anticoagulant Agents) in life-threatening hemorrhage or for urgent surgery or invasive procedures.
      a. Not indicated for the treatment of hemophilia-related bleeding because other PCC products are approved.
      b. The package insert warns of an increased thromboembolic risk with Kcentra administration. Kcentra is contraindicated in patients with known heparin-induced thrombocytopenia because the product contains heparin.
      c. Few data support the off-label use of PCCs as hemostatic agents in patients with bleeding not previously receiving anticoagulation; however, interest in this use and evidence are growing.
         i. The RETIC trial, which used PCC and concentrated fibrinogen compared with plasma in trauma patients with bleeding, was terminated early for safety concerns because of the high need of rescue therapy and massive transfusion in the plasma group.
         ii. In a double-blind, randomized, placebo-controlled superiority trial conducted in 12 French designated level I trauma centers, a total of 324 patients at risk of massive transfusion were included (PROCOAG randomized clinical trial). These patients received either intravenous 4F-PCC or a saline solution as treatment. The study's primary outcome was 24-hour all blood product consumption, and the secondary outcome was the occurrence of arterial or venous thromboembolic events. The results showed that there was no statistically or clinically significant difference in total 24-hour blood product consumption between the two groups. The 4F-PCC group consumed a median of 12 units of blood products, and the placebo group consumed a median of 11 units, with an absolute difference of 0.2 units (95% CI, -2.99 to 3.33; p=0.72). The study also showed that thromboembolic events were more common in the 4F-PCC group, with 35% of patients in this group experiencing at least one such event, compared with 24% in the placebo group. The absolute difference was 11% (95% CI, 1%–21%), with a relative risk of 1.48 (95% CI, 1.04–2.10; p=0.03). These findings do not support the routine use of 4F-PCC in such patients.
         iii. Although use of coagulation factor concentrates (CFCs) for hemostatic resuscitation is less common in the United States, the 2019 European trauma guidelines recommend CFCs as an alternative to plasma.

Table 5. Factor Content of PCCs Available in the United States

	Factor II (IU)	Factor VII (IU)	Factor IX (IU)[a]	Factor X (IU)	Additives[b]	Vial Dose (estimated factor IX IU)
3-factor PCC[c]						
Bebulin VH	100	< 5	100	100	Trace heparin	200–1200
Profilnine	150	35	100	100	N/A	500, 1000, 1500
4F-PCC[d]						
Kcentra	106.9	55.1	100	141.4	Protein C, S, Z, antithrombin III, heparin	500, 1000
aPCC[e]						
FEIBA	91.7–125	68-135	100	80–93.3	Protein C	500, 1000, 2500

[a]Expressed as international units (IU) per 100 IU of factor IX; the individual factor contents vary depending on vial size and can be determined by multiplying the individual factor content of interest by vial dose and dividing by 100 IU of factor IX equivalent.

[b]Anticoagulant factors in some PCC products are added to attenuate excessive thrombogenicity.

[c]Considered a 3-factor PCC because of limited amounts of factor VII.

[d]Considered a 4-factor PCC (4F-PCC) because of a significant concentration of factor VII.

[e]FEIBA is an aPCC that contains mainly nonactivated factors II, IX, and X; factor VII is mainly in the activated form.

FEIBA = factor eight inhibitor bypassing activity; N/A = not applicable.

Information from: Frontera JA, Lewin JJ, Rabinstein AA, et al. Guideline for reversal of antithrombotics in intracranial hemorrhage. Neurocrit Care 2016;24:6-46; FEIBA NF [prescribing information]. Baxter Healthcare, 2011.

Patient Case

Questions 3 and 4 pertain to the following case.

D.R. is a 37-year-old man with an unknown medical history who presents after a helmeted motorcycle collision into a deer. On primary survey, his endotracheal tube is secured, respiratory rate is 37 breaths/minute, SBP is 72 mm Hg, heart rate is 141 beats/minute, and Glasgow Coma Scale score is 5T. His FAST examination is positive. A massive transfusion protocol is initiated, and the patient is taken to the operating room for an emergency exploratory laparotomy.

3. Which is the best initial resuscitation strategy to manage his hypotension?
   A. Initiate a sodium chloride (0.9%) 3000-mL bolus to maintain a MAP over 65 mm Hg.
   B. Obtain an Hgb concentration to guide erythrocyte transfusions, with goal Hgb 7–9 g/dL.
   C. Initiate a norepinephrine infusion to maintain an SBP goal of 80–90 mm Hg.
   D. Initiate PBRCs, plasma, and platelets in a 1:1:1 ratio on the basis of clinical examination.

4. In the operating room, the patient is found to have grade 5 hepatic laceration and grade IV spleen injury status after perihepatic packing and splenectomy. His abdomen is left open with a wound vacuum-assisted closure, and he is taken to the ICU. On arrival to the ICU (5 hours after injury), an arterial blood gas reveals a pH 7.18, partial pressure of carbon dioxide 22 mm Hg, lactate 12.7 g/dL, and Pa_{O_2} of 86% on 40% F_{IO_2} with assist control ventilation. He has received 14 units of erythrocytes, 12 units of plasma, and 12 units of platelets. His laboratory values include aPTT 37 seconds, INR 1.3, Hgb 7.5 g/dL, and Plt 154,000/mm^3. His temperature is 99°F (37.2°C) and ionized calcium is 0.7 mmol/L. Which is the best pharmacologic treatment strategy to manage his ongoing hemorrhagic shock in a goal-directed fashion?
   A. Infuse 150 mEq of sodium bicarbonate in 1 L of sterile water at 200 mL/hour.
   B. Give 1 g of calcium chloride once infused over 1 hour.
   C. Give rFVIIa 90 mcg/kg once intravenous push.
   D. Give a tranexamic acid 1000-mg bolus and 1000 mg infused over 8 hours.

L. Reversal of Oral Anticoagulant Agents (Table 6)
   1. Patient selection
      a. Assessment of bleed severity in patients treated with oral anticoagulants is critical to guide decisions related to reversal.
      b. The American College of Cardiology (ACC) 2020 expert consensus decision pathway suggests consideration for anticoagulation reversal in a major bleed, generally defined as bleeding into a critical site, hemodynamic instability, or clinically overt bleeding.
         i. Critical bleed site:
            (a) Bleeding in a critical site can compromise the organ function, may cause severe disability, and can potentially require surgical intervention.
            (b) These sites include: intracranial, other central nervous system (spinal, ocular), thoracic (e.g., pericardial tamponade, hemothorax), intra-abdominal, retroperitoneal, airway bleeding associated with respiratory distress, or extremity bleeding (e.g., intraarticular, intramuscular) causing compartment syndrome
            (c) Intraluminal GI bleeding is not considered a critical site; however, GI bleeding may be considered a major bleed on the basis of hemodynamic or overt bleeding criteria.
         ii. Hemodynamic instability is defined as a mean arterial pressure (MAP) less than 65 mm Hg, SBP less than 90 mm Hg, SBP decrease by at least 40 mm Hg, or orthostatic blood pressure changes (SBP decrease of at least 20 mm Hg, diastolic blood pressure [DBP] decrease of at least 10 mm Hg upon standing).
         iii. Overt bleeding, with a hemoglobin decrease of at least 2 g/dL within 24 hours, or requiring the administration of at least 2 units of PRBCs, is associated with increased mortality, particularly in patients with chronic cardiovascular disease.
         iv. Recommendations for managing a major bleed:
            (a) Discontinue the anticoagulant, provide manual compression, provide supportive care, assess and manage comorbidities that could contribute to bleeding (e.g., thrombocytopenia, uremia, liver disease, antiplatelets), and consider surgical or procedural management of the bleeding site.
            (b) Specific anticoagulation reversal is indicated for bleeds at a critical site or those causing hemodynamic instability. For overt bleeding, initial bleeding control procedures are recommended, and reversal is indicated if the earlier measures fail to control the bleed.
   2. Warfarin
      a. Warfarin has a long physiologic half-life; thus, the reversal decision is usually based on the presence of an elevated INR in a life-threatening hemorrhage or the need for urgent surgery or an invasive procedure within 24 hours.
      b. Phytonadione (vitamin K) administration is the definitive reversal of warfarin, promoting hepatic production of vitamin K–dependent clotting factors depleted by warfarin.
         i. In life-threatening hemorrhage, injectable phytonadione at a dose of 5–10 mg given intravenously is preferred.
         ii. Previous reports of anaphylaxis are related to solubilizing agents no longer included in the formulation.
         iii. Phytonadione has a delayed onset (6–12 hours); therefore, clotting factors are given concurrently to expedite reversal.
      c. Clotting factors can be given as either plasma or PCC.
         i. Plasma has traditionally been used, but it requires large volumes, leads to incomplete INR correction, requires compatibility testing, and requires an extended time to achieve hemostasis.

ii. Evidence to support 4F-PCC over plasma:
   (a) Compared with plasma in life-threatening hemorrhage, 4F-PCC is superior for laboratory reversal, is noninferior for clinical hemostasis, and has a similar rate of thromboembolic events.
   (b) In an urgent surgical or invasive procedure, 4F-PCC is superior for rapid INR reversal and achievement of hemostasis.
   (c) An integrated analysis of both life-threatening hemorrhage and urgent surgical trials suggests similar rates of thromboembolic events (4F-PCC 7.3% vs. fresh frozen plasma 7.1%), with fluid overload more common in the plasma group (4F-PCC 12.7% vs. fresh frozen plasma 4.7%). Although nonactivated 4F-PCC has a black box warning for an increased risk of arterial and venous thromboembolic events, this risk appears to be similar to that of plasma.
   (d) Although 4F-PCC is associated with a high direct acquisition cost, pharmacoeconomic analyses are investigating similar strategies for warfarin reversal to fully understand the cost implications.

iii. rFVIIa is no longer recommended for warfarin reversal because of incomplete correction of factor concentrations inhibited by warfarin.

iv. Recommendations in a life-threatening hemorrhage:
   (a) Phytonadione 10 mg intravenously and a 4F-PCC (dose is determined by patient weight and pretreatment INR; see Table 7) are preferred for most patients.
   (b) Fixed dose 4F-PCC (e.g., 1000–1500 units) has been evaluated for warfarin reversal showing correction of the INR, but the lower dose may fail to correct the INR to goal, necessitating additional treatment. The 2020 ACC guidelines consider fixed low dose (1000 units for non-intracranial major bleed and 1500 units for intracerebral hemorrhage) as an alternative to the package insert dosing, but this is off-label and has lower-quality supporting evidence.
   (c) In patients with a contraindication to 4F-PCC (e.g., a history of heparin-induced thrombocytopenia), fresh frozen plasma 10–15 mL/kg or non-heparin containing PCC may replace 4F-PCC for reversal.
   (d) Although the goal INR is controversial, the 2016 Neurocritical Care Society guidelines recommend trending an INR in patients with intracranial hemorrhage with repeat reversal strategies until the INR is less than 1.4. Randomized controlled trials of 4F-PCC included only patients with an INR of 2 or greater.

3. Direct oral anticoagulants (DOACs) – Dabigatran/apixaban/rivaroxaban/edoxaban:
   a. Coagulation assays that have direct, linear relationships to the DOAC anticoagulation activity are not available for clinical use (Table 6).
   b. Therefore, the initial approach to reversal should be guided by clinical bleeding and the knowledge of preexisting DOAC use rather than coagulation assays.
   c. Given the relatively short DOAC half-life in patients with normal end-organ function, establishing the time from the last dose is critical in determining whether the drug is contributing to bleeding. In general, if it has been more than 3–5 half-lives since the last dose, reversal is probably not indicated (Table 6).

Table 6. Summary of Oral Anticoagulants

	Warfarin	Dabigatran	Apixaban	Rivaroxaban	Edoxaban
Action	Vitamin K antagonist	Direct factor IIa inhibitor	Direct factor Xa inhibitor	Direct factor Xa inhibitor	Direct factor Xa inhibitor
Peak action	4–5 days	~2 hr	~2 hr	~2 hr	1–2 hr
Half-life (hr) CrCl: > 80 CrCl: 50–79 CrCl: 30–49 CrCl < 30	> 48 hr	 12–14 17 19 28	 5–9 9 9 10	 8–15 15 18 17	 8.5 9 9 9.5
Clinical coagulation monitoring	Protime/INR: Quantitative	aPTT and TT: Qualitative ECT and dilute TT: Quantitative	Protime Rivaroxaban, edoxaban: Qualitative, if validated Apixaban: Insensitive Chromogenic anti-Xa: Quantitative Heparin or low-molecular-weight heparin assay: Qualitative		

anti-Xa = anti-factor Xa; CrCl = creatinine clearance reported as mL/min/1.73 m^2; ECT = ecarin clotting time; TT = thrombin time.
Information from: Dager WE. Developing a management plan for oral anticoagulation reversal. Am J Health Syst Pharm 2013;70:S21-31.

d. Activated charcoal may be considered if the last dose was administered less than 2 hours previously. Given significant enterohepatic recirculation, charcoal can be given up to 6 hours after the last apixaban dose.
e. Dabigatran reversal: Idarucizumab is FDA approved for dabigatran reversal.
    i. Idarucizumab is a monoclonal antibody that directly binds both bound and unbound dabigatran with greater than 350 times the affinity of thrombin.
    ii. In the full analysis of 503 patients with life-threatening hemorrhage or need for urgent procedure, all who received idarucizumab 5 g intravenously had maximum dilute TT reversal by 4 hours after idarucizumab administration. Given that dabigatran has a volume of distribution of greater than 50 L, about 23% of patients developed redistribution of unbound dabigatran and associated coagulopathy at 12–24 hours. The efficacy of repeat doses in this setting has not been established. The package insert recommends that if there is clinically relevant bleeding (or need for a second emergency surgery/urgent procedure) in combination with elevated coagulation parameters (e.g., aPTT, ecarin clotting time), an additional 5-g dose of idarucizumab may be considered. Therefore, an aPTT should be obtained to rule out the presence of dabigatran before a repeat dose is given.
    iii. The thrombotic event rate at 90 days was 6.8%. All thrombotic events that occurred within 72 hours were in patients for whom anticoagulation was not reinitiated (also because of half-life of 45 minutes).
    iv. Given the availability of idarucizumab, adjunctive reversal options such as dialysis should only be considered if there are concurrent indications on the basis of renal function or if idarucizumab is ineffective.
f. Factor Xa inhibitors: Andexanet alfa is FDA approved for rivaroxaban and apixaban reversal in life-threatening hemorrhage.
    i. Andexanet alfa is a modified factor Xa decoy protein such that it directly binds to the factor Xa inhibitors in a 1:1 ratio to inactivate their anticoagulant response.
        (a) Andexanet alfa structural modifications render it incapable of having prothrombotic or antithrombotic effects.

(b) Mutation of the active serine site ensures that andexanet lacks the catalytic activity needed to convert prothrombin to thrombin.
(c) Deletion of the GLA fragment makes andexanet unable to bind to phospholipids and interact with factor V and calcium to form a prothrombinase complex.

ii. Andexanet alfa reverses the effects of both direct (DOAC) and indirect (low-molecular-weight heparin) factor Xa inhibitors.

iii. The ANNEXA-4 trial evaluated the effects of andexanet alfa in patients with acute major bleeding and recent ingestion of factor Xa inhibitors, predominantly apixaban and rivaroxaban.
(a) After an initial bolus of andexanet, there was a 90%–92% reduction in the anti-factor Xa (anti-Xa) activity from baseline. This was sustained through the 2-hour infusion of the study drug, allowing for a hemostatic plug to form.
(b) Of note, only 72% of patients met the predefined threshold for anticoagulation activity (baseline anti-Xa concentrations of 75 ng/mL or greater for DOACs or 0.25 IU/mL or greater for enoxaparin) to be included in this efficacy analysis.
(c) An important limitation of this study was the exclusion of patients expected to require surgery within 12 hours, a population commonly encountered when treating bleeding patients.

iv. Andexanet alfa has a short half-life (1 hour); thus, redistribution and associated anticoagulant activity are also common after discontinuing andexanet.
(a) In theory, this 2-hour time interval allows for a hemostatic plug to form and bleeding to stop.
(b) In the ANNEXA-4 trial, anti-Xa activity rebounded. Two hours after discontinuing the infusion, anti-Xa activity decreased only 32%–42% relative to baseline.
(c) However, at 12 hours, 82% of patients met the co-primary end point of good or excellent clinical hemostasis. This brings into question the clinical significance of the rebound effect. Further research is needed to determine the importance of this rebound effect in patients with ongoing bleeding, patients with critical site bleeds, or surgical populations.

v. Anticipated limitations when using andexanet in clinical practice:
(a) The largest limitation of the ANNEXA-4 trial is the lack of a control group, with little ability to determine how andexanet affects patient-centered outcomes.
(b) Given that only 72% of patients were eligible for the efficacy analysis, these results may have an exaggerated effect relative to that of the general population.
(c) One of the greatest concerns is the relative short duration of laboratory reversal, given the short half-life of andexanet alfa and its reversible binding with anti-Xa inhibitors.
(d) Implications are unclear for patients who continue to have clinical evidence of bleeding beyond the andexanet infusion and the need to repeat doses or extend infusions.
(e) Notable exclusion criteria from this trial include requirement of surgery as well as severe brain injury (average GCS score was 14). Implications of short-term reversal may be even greater for these high-acuity patient populations.

vi. Safety of andexanet alfa:
(a) 10% of the population had a thrombotic event, with most occurring before the initiation of therapeutic anticoagulation.
(b) Andexanet alfa does bind to tissue factor pathway inhibitor, an endogenous factor Xa inhibitor.
(c) This rate is generally higher than what has been shown for other reversal agents in bleeding patients, including PCCs for warfarin reversal (7%–8%) and idarucizumab for dabigatran reversal (6.35%–7.4%).

(d) It is important to recognize differences in population risks that confound cross-study comparisons, given that the updated ANNEXA-4 analysis included at least double the population with intracranial hemorrhage compared with other reversal studies. In this setting, reintroduction of anticoagulation may be delayed.

vii. Andexanet alfa dosing is complex (Table 7) and individualized according to the underlying anticoagulant and time from last dose. For example, rivaroxaban achieves higher serum concentrations, so andexanet doses are higher for rivaroxaban with recent administration than for apixaban.

viii. Although andexanet alfa may work for all direct and indirect anti-Xa inhibitors, it is only FDA approved for apixaban and rivaroxaban reversal in life-threatening or uncontrolled bleeding.

(a) Use for other anticoagulant-related bleeding is off-label, with overall lower supporting evidence.

(b) If used for edoxaban or enoxaparin, the dose under investigation is high dose.

g. The antidotes for DOAC reversal were FDA reviewed under the Fast Track designation.

i. Because it is unethical to randomize a patient to placebo during antidote trials, the studies establishing efficacy and safety are single-arm trials.

ii. The FDA has requested an andexanet alfa arm compared with a usual care arm for anti-Xa reversal in patients with intracranial hemorrhage. The ANNEXA-I study (NCT03661528) aimed to evaluate the effectiveness of andexanet alfa compared with usual care in treating intracranial hemorrhage. Although the full results of the trial have not been reported, preliminary findings presented at conferences indicate the study met its prespecified criteria for hemostatic efficacy when compared with standard care. This led to early termination of the study based on a planned interim analysis. However, the study also found that andexanet alfa was associated with a higher incidence of thrombotic events, including ischemic stroke and myocardial infarction, than usual care.

iii. Therefore, it remains unknown whether the novel antidotes will affect morbidity or mortality associated with anticoagulants in life-threatening hemorrhage.

h. PCCs for DOAC reversal

i. PCCs and aPCCs have traditionally been used off-label as a nonspecific reversal agent for DOAC reversal before the development of novel antidotes.

ii. Supporting data are primarily based on laboratory reversal of factor Xa inhibitors from in vitro animal models, ex vivo human samples, and healthy volunteer subjects. These trials have important limitations and inconsistent results.

iii. More recently, PCC was an effective reversal for anti-Xa inhibitors in bleeding patients, as evaluated by hemostatic response at defined time intervals, but these trials are limited by single-arm observational study design and lack of analysis of laboratory reversal.

iv. PCC has not consistently been shown to reverse laboratory values associated with dabigatran.

v. Several guidelines recommended 4F-PCC or aPCC administration at 25–50 units/kg for anti-Xa reversal before the availability of andexanet alfa. Guidelines may continue to evolve as more robust data comparing andexanet alfa with usual care become available. Clinicians should carefully consider the non-sustained duration of reversal of andexanet alfa and the lack of data in emergency surgical procedures.

Table 7. Anticoagulant Reversal Agent Dosing in Life-Threatening Hemorrhage

<table>
<tr><th colspan="2">Which Anticoagulant</th><th>Reversal Agent</th><th>Dose</th></tr>
<tr><td colspan="2">Warfarin with elevated INR</td><td>4F-PCC</td><td>INR 2 to < 4: 25 units/kg (max 2500 units)
INR 4–6: 35 units/kg (max 3500 units)
INR > 6: 50 units/kg (max 5000 units)</td></tr>
<tr><td colspan="2">Dabigatran within 3–5 expected half-lives</td><td>Idarucizumab</td><td>5 g IV push once
Limited data to repeat 5 g</td></tr>
<tr><td colspan="4">Factor Xa inhibitor within 3–5 expected half-lives</td></tr>
<tr><td>Agent</td><td>Last Dose</td><td rowspan="3">Andexanet alfa</td><td rowspan="3">Low dose: 400-mg bolus at a target rate of 30 mg/min, 4 mg/min for 120 min
High dose: 800-mg bolus at a target rate of 30 mg/min and 8 mg/min for 120 min
If time from last dose not known, use appropriate dose if consumed within 8 hr</td></tr>
<tr><td>Rivaroxaban</td><td>≤ 10 mg: Low dose
> 10 mg within 8 hr: High dose
> 10 mg after ≥ 8 hr: Low dose</td></tr>
<tr><td>Apixaban</td><td>≤ 5 mg: Low dose
> 5 mg within 8 hr: High dose
> 5 mg after ≥ 8 hr: low dose</td></tr>
<tr><td>Edoxaban
Enoxaparin</td><td>High dose at any time
Off-label</td><td></td><td></td></tr>
<tr><th colspan="2">Which Anticoagulant</th><th>Reversal Agent</th><th>Dose</th></tr>
<tr><td colspan="2">Direct and indirect anti-Xa inhibitors</td><td>4F-PCC
aPCC
Off-label</td><td>25–50 units/kg</td></tr>
</table>

IV = intravenous(ly).

Information from: Sarode R, Milling TJ, Refaai MA, et al. Efficacy and safety of 4-factor prothrombin complex concentrate in patients on vitamin K antagonists presenting with major bleeding: a randomized, plasma-controlled, phase IIIb study. Circulation 2013;128:1234-43; Pollack CV, Reilly PA, Eikelboom J, et al. Idarucizumab for dabigatran reversal. N Engl J Med 2015;373:511-20; and Connolly SJ, Milling TJ, Eikelboom JW, et al. Andexanet alfa for acute major bleeding associated with factor Xa inhibitors. N Engl J Med 2016;365:1131-41.

M. Reversal of Antiplatelet Agents (Table 8)
   1. Controversies remain regarding the impact of antiplatelet agents on traumatic bleeding.
   2. The PATCH trial was the first randomized controlled trial evaluating the impact of platelet transfusions for spontaneous intracranial hemorrhage in patients receiving antiplatelet therapy for at least 7 days prior.
      a. Odds of death were higher in the platelet transfusion group at 3 months than in placebo (adjusted OR 2.05; 95% CI, 1.18–3.56; p=0.0114), suggesting platelet transfusions are inferior to standard of care in non-operative patients.
      b. Study limitations include the lack of platelet function tests, inability to ensure medication adherence before the event, exclusion of surgical patients, and predominant use (greater than 90%) of aspirin as the antiplatelet agent.
      c. How this study translates to traumatic hemorrhage is unclear, and randomized controlled trials are needed.
   3. The 2019 European guidelines for managing traumatic bleeding recommend platelet transfusions in patients who have been treated with antiplatelet agents in the following scenarios:
      a. Continued bleeding with documented platelet dysfunction (grade 2C)
      b. Intracranial hemorrhage undergoing surgery (grade 2B)

4. Desmopressin enhances platelet adherence and aggregation.
    a. Although its clinical efficacy is controversial, desmopressin 0.3–0.4 mcg/kg intravenously may be considered to reverse platelet-inhibiting drugs or von Willebrand disease but may not be routinely used in trauma patients with bleeding. If administering desmopressin, the 2019 European trauma guidelines recommend administering an antifibrinolytic (e.g., tranexamic acid) in parallel.
    b. There is a low risk of serious adverse effects, including hypervolemia, hyponatremia, facial flushing, and rare thrombosis.

Table 8. Antiplatelet Reversal in Life-Threatening Hemorrhage

Generic Drug (trade name): Mechanism of Action	Coagulation Assay Effects	Usual Half-Life (hr)	Duration of Platelet Inhibition	Recommendations
Aspirin Inhibits platelet activation Irreversible inhibition of COX-1 and COX-2 enzymes	Optimal platelet function assay and thresholds controversial Light transmission platelet aggregation is gold standard of platelet function testing but is not routinely available TEG with platelet mapping, VerifyNow, platelet function analyzer, and bleeding time may be used	3	Lifetime of platelets 5–7 days	Use caution with antiplatelet reversal in patients with active or recent (within 3 mo) coronary artery disease <u>Reversal strategies:</u> **Platelets*** Neurosurgical hemorrhage: 1 apheresis unit (moderate quality of evidence) Neurosurgery is not indicated: Platelet transfusions controversial and not recommended by the Neurocritical Care Society (low quality of evidence) -and/or- **Desmopressin (DDAVP) 0.4 mcg/kg IV push once** (low quality of evidence) *Transfused platelets with ticagrelor may become inhibited because of reversible activity at the P2Y12 receptor
Dipyridamole ± aspirin (Aggrenox, Persantine) Inhibits platelet activation Increases endogenous adenosine		10–12	2–3 days (dipyridamole) Lifetime of platelets 5–7 days for aspirin	
Clopidogrel (Plavix) Inhibits platelet activation Active metabolite irreversibly blocks ADP		6	Lifetime of platelets 5–7 days	
Prasugrel (Effient) Inhibits platelet activation Active metabolite irreversibly blocks ADP		2–15	Lifetime of platelets 5–7 days	
Ticagrelor (Brilinta) Inhibits platelet activation Reversibly blocks ADP		7–9	3–5 days	

AA = arachidonic acid; ADP = adenosine diphosphate; COX = cyclooxygenase.

Information from: Levi M, Eerenberg E, Kamphuisen PW, et al. Bleeding risk and reversal strategies for old and new anticoagulants and antiplatelet agents. J Thromb Haemost 2011;9:1705-12; Mega J, Simon T. Pharmacology of antithrombotic drugs: an assessment of oral antiplatelet and anticoagulant treatments. Lancet 2015;386:281-91; Spahn D, Bouillon B, Cerny V, et al. The European guideline on management of major bleeding and coagulopathy following trauma: fifth edition. Crit Care 2019;23:98; Frontera JA, Lewin JJ, Rabinstein AA, et al. Guideline for reversal of antithrombotics in intracranial hemorrhage. Neurocrit Care 2016;24:6-46.

Patient Cases

5. A 54-year-old man with an unknown medical history presents after a rollover automobile accident. On primary survey, his respiratory rate is 34 breaths/minute, SBP is 79 mm Hg, and Glasgow Coma Scale score is 3. The FAST examination is positive, a massive transfusion protocol is initiated, and the patient is taken to the operating room for an emergency exploratory laparotomy. In the operating room, he is found to have grade 5 splenic injury, so the spleen is removed. Four hours after admission, he has received 14 units of PRBCs, 12 units of fresh frozen plasma, and 10 units of platelets. His laboratory values include aPTT 66 seconds, INR 1.7, fibrinogen 176 mg/dL, Plt 160,000/mm^3, and ionized calcium 1.1 mmol/L. The TEG is normal except for an LY30 of 0% (normal 0.8%–8%). The patient's wife arrives at the hospital and clarifies that he was out for a Sunday afternoon ride. She also shares that the patient has a history of atrial fibrillation and takes dabigatran (last dose this morning). Which is the best pharmacologic treatment of his ongoing hemorrhagic shock?
   A. Administer 4F-PCC 50 units/kg.
   B. Administer a tranexamic acid 1-g bolus with 1-g infusion over 8 hours.
   C. Administer an idarucizumab 5-g intravenous push.
   D. Administer an idarucizumab 5-g intravenous push and a tranexamic acid 1-g bolus, followed by 1 g infused over 8 hours.

6. B.P is a 76-year-old man (120 kg) with atrial fibrillation on warfarin admitted to the ED with weakness and hematemesis. Pertinent vital signs on admission are as follows: blood pressure 79/52 mm Hg, heart rate 136 beats/minute with a rhythm of sinus tachycardia, and respiratory rate 26 breaths/minute. Frank red blood is noted on nasogastric lavage. On physical examination, the patient is confused, with clammy extremities. Pertinent laboratory values are as follows: Hgb 6.3 g/dL, INR 9.2, and Plt 300,000/mm^3. In addition to intravenous phytonadione, which therapy is most appropriate for managing warfarin reversal in this patient?
   A. 4F-PCC 1000 units.
   B. 4F-PCC 5000 units.
   C. rFVIIa 1-mg intravenous push.
   D. 4F-PCC 6000 units.

II. OBSTRUCTIVE SHOCK

A. Etiology and Epidemiology
   1. Obstructive shock occurs because of extracardiac obstruction to flow in the cardiovascular system.
   2. The source of extracardiac obstruction may be either impaired diastolic filling (e.g., cardiac tamponade, tension pneumothorax, or constrictive pericarditis) or impaired systolic contraction (e.g., massive PE, acute or chronic PH or aortic dissection).
   3. Obstructive shock is relatively rare, and only about 2% of patients requiring vasoactive medications have this type of shock.

B. Pathophysiology
   1. The pathophysiologic hemodynamic hallmark of obstructive shock is decreased CO.
   2. The specific pathophysiologic derangements depend on the underlying cause of the extracardiac obstruction but are generalized as follows:

a. In impaired diastolic filling, RV preload is significantly decreased because of the inhibition of venous return.
   i. In cardiac tamponade, an accumulation of fluid in the pericardium leads to an increase in pericardial pressure.
   ii. This increases and equalizes diastolic pressures between the left and right heart (equalization of central venous pressure [CVP], pulmonary artery diastolic pressure, and pulmonary capillary wedge pressure [PCWP] and impaired ventricular filling). CVP and PCWP elevations should not be mistaken as representing an increase in ventricular volume (preload).

b. In impaired systolic function, ventricular afterload is acutely increased, leading to ventricular failure.
   i. This typically occurs in an acute RV afterload (pulmonary vascular resistance) increase caused by a massive PE or in acute PH.
   ii. An acute increase in LV afterload does not typically lead to shock.
   iii. An acute rise in RV afterload leads to reduced RV CO and a subsequent decrease in LV CO. Subsequently, systemic hypotension develops, leading to reduced RV tissue perfusion (decreased right coronary artery perfusion), RV free wall ischemia, reduced RV free wall contractility, and further impairment of RV CO (a vicious cycle).
   iv. In addition, acute RV pressure overload leads to a shift of the intraventricular septum toward the LV, impairing LV diastolic filling (because of intraventricular dependence) and further decreasing in LV CO.

C. Resuscitation and Treatment
1. Fluid administration and vasoactive medications may be used as a temporizing measure to increase tissue perfusion.
   a. Intravenous fluids (typically crystalloids) are usually recommended, but they may not improve CO.
      i. Cardiac tamponade: Patients with preexisting hypovolemia may respond to fluids, but in general, hemodynamics may not improve with fluid administration. Despite this finding, fluid administration is usually recommended in cardiac tamponade.
      ii. Massive PE: Initial fluid administration improves CO, but care should be taken because excessive fluid administration can lead to further RV dilation and impaired LV CO from worsened septal shifting and decreased LV filling (because of intraventricular dependence).
      iii. Acute or chronic PH: Optimization of fluids in patients with acute or chronic PH is challenging. Some patients, such as those with signs of intravascular volume depletion, may require fluids. Others may require diuretics to reduce RV dilation and improve LV filling, even with vasoactive medication administration.
   b. Vasopressors should be initiated to increase MAP and maintain an adequate perfusion pressure.
      i. This is particularly important in a massive PE because adequate right coronary artery perfusion is of greatest importance to prevent/reduce RV free wall ischemia.
      ii. Caution must be used because catecholamine vasopressors may increase pulmonary vascular resistance, which may worsen RV dysfunction.
   c. Inotropes may increase RV CO in a massive PE or acute or chronic PH but are likely ineffective in tamponade.
   d. Inhaled nitric oxide or aerosolized prostacyclin therapy may decrease RV afterload in acute or chronic PH, but neither therapy is likely effective for massive PE or cardiac tamponade.
2. Definitive treatment of the extracardiac obstruction is paramount.
   a. Impaired diastolic filling
      i. Cardiac tamponade: Pericardiocentesis or surgical evacuation and potential drain placement
      ii. Tension pneumothorax: Needle decompression and potential chest tube thoracostomy

b. Impaired systolic function (massive PE): Embolism dissolution (thrombolytic therapy) or removal (surgical or catheter thrombectomy)
   i. Thrombolytic agents (Table 9) bind to the plasminogen/plasmin complex, which may be either circulating or bound to the clot surface.
   ii. This binding hydrolyzes a peptide bond to form free plasmin. Circulating plasmin is quickly neutralized by α-antiplasmin, but fibrin-bound plasmin subsequently hydrolyzes bonds within the clot matrix, leading to clot lysis.
   iii. As such, thrombolytic agents can lead to rapid PE dissolution and a subsequent decrease in RV afterload (pulmonary vascular resistance), but they may also cause bleeding.

Table 9. Thrombolytic Agents for Acute PE

Drug	Dose		Pearl
	Initial	Infusion	
Alteplase[a]	10 mg over 10 min	90 mg over 2 hr	Fibrin selective half-life: 5 min Dose in cardiac arrest may be 50 mg
	Total dose: 100 mg infused over 2 hr		
Reteplase[b]	Two doses of 10 units IV push over 2 min, given 30 min apart		Fibrin selective half-life: 13–16 min
Tenecteplase[b]	< 60 kg: 30-mg push over 5–10 s 61–70 kg: 35-mg push 71–80 kg: 40-mg push 81–90 kg: 45-mg push > 91 kg: 50-mg push	N/A	Fibrin selective (14-fold compared with alteplase) half-life: 90–130 min; only need bolus dose Less expensive than alteplase
Streptokinase[a]	250,000 units over 30 min	100,000 units/hr over 24 hr	Nonselective Antigenic
Urokinase[a]	4400 units/kg over 10 min	4400 units/kg/hr over 12 hr	Nonselective

[a]Indicates FDA approved for the treatment of PE.

Indicates drug is not FDA approved for the treatment of PE.

PE = pulmonary embolism.

Information modified from: Daley MJ, Lat I. Clinical controversies in thrombolytic therapy for the management of acute pulmonary embolism. Pharmacotherapy 2012;32:158-72.

   iv. In a 2004 meta-analysis, thrombolytic agents were not shown to decrease mortality when evaluated among all patients with a PE compared with heparin alone (6.7% vs. 9.6%; OR 0.67 [95% CI, 0.40–1.12]), but they may improve outcomes in patients with an increased risk of death. As such, early PE-related mortality risk stratification is necessary to guide thrombolytic therapy administration.
      (a) High-risk or massive PE. Defined by the 2011 American Heart Association (AHA) guidelines not by clot burden, but by acute PE causing hemodynamic changes: Hypotension (SBP less than 90 mm Hg for at least 15 minutes or requiring vasoactive support not from another cause), pulselessness, or bradycardia (heart rate less than 40 beats/minute with signs of shock). The 2019 European Society of Cardiology (ESC) and European Respiratory Society (ERS) guidelines for acute PE define hemodynamic instability as including one of the following presentations:
         (1) Cardiac arrest,
         (2) Obstructive shock (SBP less than 90 mm Hg or vasopressors required to achieve SBP of 90 mm Hg or greater despite adequate filling status AND end-organ hypoperfusion), or

(3) Persistent hypotension (SBP less than 90 mm Hg or SBP decrease of 40 mm Hg or greater for more than 15 minutes, not caused by new-onset arrhythmia, hypovolemia, or sepsis)

(b) Intermediate-risk or submassive PE.

(1) AHA 2011 classification: Acute PE without systemic hypotension but with either (1) RV dysfunction (RV dilation or systolic dysfunction on echocardiogram, RV dilation on chest CT, brain natriuretic peptide greater than 90 pg/mL, N-terminal pro-brain natriuretic peptide greater than 500 pg/mL, or electrocardiographic changes [new complete or incomplete right bundle-branch block, anteroseptal ST-segment elevation or depression, or anteroseptal T-wave inversion]) or (2) myocardial necrosis (troponin I greater than 0.4 ng/mL or troponin T greater than 0.1 ng/mL) (2) ESC/ERS 2019 classification: Simplified Pulmonary Embolism Severity Index score of 1 or greater (i.e., age older than 80; cancer, chronic respiratory disease, or cardiac disease; heart rate 110 beats/minute or greater; SBP less than 100 mm Hg; or Sa_{O_2} less than 90%), regardless of whether there is RV strain

(c) Intermediate-risk PE may further be classified by the presence of both RV dysfunction and myocardial necrosis, termed *intermediate-high risk*. If only RV dysfunction or myocardial necrosis is present, it is classified as *intermediate-low risk*.

(d) Low-risk PE: Acute PE not meeting the definition of massive or submassive PE

v. Thrombolytics in cardiac arrest:

(a) The 2020 AHA International Consensus on Cardiopulmonary Resuscitation and Emergency Cardiovascular Care (Circulation 2020;142:S92-S139) remains unchanged from the 2015 AHA guidelines, stating there is no consensus on the ideal thrombolytic dose in PE-associated cardiac arrest. Contemporary examples of dosing regimens include either an alteplase bolus of 50 mg (repeated if needed to complete a 100-mg intravenous total dose) or weight-based tenecteplase if PE is confirmed as the contributing cause of the arrest.

(b) The 2020 AHA guidelines have a weak recommendation to administer fibrinolytic drugs for cardiac arrest when PE is the suspected cause of cardiac arrest (very low-certainty evidence). When PE is the confirmed cause of cardiac arrest, fibrinolytic drugs or surgical embolectomy or percutaneous mechanical thrombectomy may be used (very low-certainty evidence). The task force considered increased bleeding risk from fibrinolytics administered to patients who may not have PE. In the TROICA study, the largest study of thrombolysis during cardiac arrest, thrombolysis was associated with increased bleeding risk but no difference in major bleeding complications. The task force concluded that patients are far more likely to die from cardiac arrest than from treatment.

(c) These recommendations are based on possible improved outcomes with thrombolytics in peri-arrest settings, but prospective evaluations are conflicting.

vi. Thrombolytics in massive PE:

(a) The 2021 update to the CHEST guidelines and the 2019 ESC/ERS guidelines recommend systemic thrombolytics for patients with a massive/high-risk PE and an acceptable risk of bleeding complications.

(b) These recommendations are supported by a meta-analysis, which found that in patients with a massive PE, thrombolytics were associated with a lower rate of recurrent PE or death than heparin alone (9.4% vs. 19.0%; OR 0.45 [95% CI, 0.22–0.92]).

(c) A 2012 analysis of a National Inpatient Sample containing over 70,000 patients with an unstable PE (defined as the presence of shock or a requirement for mechanical ventilation) suggested reduced mortality attributable to PE with the use of systemic thrombolytics (8.4% vs. 42%; $p<0.0001$).

(d) Systemic thrombolytics for a massive PE are likely underused because about 70% of patients in this database were not treated.

vii. Use of thrombolytics in submassive PE is controversial and should be based on the risk-benefit profile:
   (a) In a study of patients with a submassive PE, adding alteplase 100 mg infused over 2 hours to heparin compared with heparin plus placebo (heparin alone) was associated with a lower rate of death or clinical deterioration requiring an escalation in treatment (11.0% vs. 24.6%, p=0.006). This result was driven by more patients in the heparin-alone arm who received secondary thrombolytics (23.2% vs. 7.6%, p=0.001), which may have been influenced by the investigators' ability to break the blinding of treatment allocation in the study. Bleeding did not differ between study arms.
   (b) The PEITHO study evaluated patients with a submassive PE (fulfilling both RV dysfunction and myocardial necrosis criteria) randomized to weight-based tenecteplase plus heparin or placebo plus heparin (heparin alone).
      (1) Between randomization and day 7, patients allocated to tenecteplase less commonly died or had hemodynamic decompensation (2.6% vs. 5.6%, p=0.02; number needed to treat 33 patients [95% CI, 18–162 patients]) but more commonly had major extracranial bleeding (6.3% vs. 1.2%, p<0.001; number needed to harm 20 patients [95% CI, 13–35 patients]) and stroke (2.4% vs. 0.2%, p=0.003; number needed to harm 47 patients [95% CI, 26–107 patients]).
      (2) The difference in stroke incidence was driven by a higher incidence of hemorrhagic stroke in the tenecteplase arm (2.0% vs. 0.2%, p=0.01; number needed to harm 57 patients [95% CI, 30–164 patients]).
      (3) The overlapping 95% CIs for number needed to treat and number needed to harm for clinically important outcomes call into question the risk-benefit profile of tenecteplase for a submassive PE.
   (c) Around 70% of patients from the PEITHO trial had a long-term follow-up for at least 24 months (average 37.8 months).
      (1) Overall and cause-specific mortality at 30 days or long term did not differ in those who received tenecteplase compared with placebo.
      (2) Of the 290 patients who had echocardiograms with long-term follow-up, CTEPH was confirmed in 2.1% of those who received tenecteplase compared with 3.2% who did not (p=0.79).
      (3) In addition, functional status, symptoms, RV dysfunction, and survival by day 30 did not differ between tenecteplase and placebo.
   (d) As a result of this trial, the 2021 CHEST guidelines and the 2019 ESC/ERS guidelines recommend against routine use of systemic thrombolytics in intermediate- or low-risk PE. However, in select patients with signs or risk of deterioration and a low bleeding risk, systemic thrombolytics may be considered.

viii. The risk-benefit of thrombolytics is best determined on a case-by-case basis by the bedside clinician (Table 10).
   (a) In the PEITHO trial, a subgroup analysis showed that age older than 75 may be associated with a risk of major extracranial bleed (OR 20.38; 95% CI, 2.69–154.53; p=0.09).
   (b) Commonly used bleeding scores such as HAS-BLED moderately predict the risk of an intracranial hemorrhage after thrombolytic agents for PE but lack precision.

ix. In extreme circumstances (e.g., massive PE with impending cardiac arrest), even the presence of strong risk factors for bleeding may not preclude some clinicians from administering thrombolytics because of the potential for benefit from the therapy. In these settings, even the "contraindications" noted in the product labeling, other than active internal bleeding, may be considered "relative contraindications" by some clinicians for thrombolytic administration.

Table 10. Risk Factors for Bleeding After Thrombolytic Therapy

Alteplase package insert contraindications	
Active internal bleeding	Recent (within 3 mo) intracranial or intraspinal surgery or serious head trauma
Current cerebrovascular accident (intracranial hemorrhage or ischemic stroke)	Current severe uncontrolled hypertension[a]
Malignant intracranial neoplasm, arteriovenous malformation, or aneurysm	Known bleeding diathesis[b]
Alteplase package insert warnings	
Recent major surgery	Significant closed-head or facial trauma with radiographic evidence of bony fracture or brain injury within 3 mo
Cerebrovascular disease	Recent GI or genitourinary bleeding
Hemostatic defects (including those secondary to severe hepatic or renal disease)	Hypertension (SBP ≥ 175 mm Hg and/or DBP ≥ 110 mm Hg)
Acute pericarditis	Subacute bacterial endocarditis
Recent trauma	Significant hepatic dysfunction
Age > 75	Needle puncture at noncompressible site
Left heart thrombus	Pregnancy
Septic thrombophlebitis or occluded arteriovenous cannula at seriously infected site	Diabetic hemorrhagic retinopathy or other hemorrhagic ophthalmic conditions
Recent receipt of oral anticoagulants	
Bleeding risk factors reported in published literature	
Recent internal bleeding	Suspected aortic dissection
Poorly controlled baseline hypertension[a]	Acute pancreatitis
Acute myocardial infarction	Cardiopulmonary resuscitation > 10 min
Stool occult blood positive	Bilirubin concentration > 3 mg/dL
Presence of an intra-aortic balloon pump	Dementia
African American race	Coagulopathy[b]

[a]Undefined for PE, but noted as > 185 mm Hg systolic or > 110 mm Hg diastolic for acute ischemic stroke.

[b]Undefined for PE, but noted as INR > 1.7 or Plt < 100,000/mm3 for acute ischemic stroke.

Adapted with permission from: Curtis GC, Lam SW, Reddy AJ, et al. Risk factors associated with bleeding after alteplase administration for pulmonary embolism: a case-control study. Pharmacotherapy 2014;34:818-25; and Daley MJ, Lat I. Clinical controversies in thrombolytic therapy for the management of acute pulmonary embolism. Pharmacotherapy 2012;32:158-72.

x. Catheter-directed thrombolysis (CDT) is an emerging treatment that enables the administration of lower-dose thrombolytic agents directly into the thrombus.
   (a) CDT has the advantage of concurrent mechanical disruption, including ultrasound-assisted fragmentation, aspiration, or embolectomy.
   (b) Applying CDT, several clinical trials have shown improved hemodynamics and strain on the RV with minimal risk of a major bleed (less than 1%).
   (c) However, prospective comparative studies evaluating long-term outcomes with CDT are currently lacking, and the site experiences and resources required limit widespread use.

(d) The 2021 CHEST guidelines recommend systemic thrombolytic therapy over CDT for most patients; however, CDT may be considered in patients with acute PE associated with hypotension who also have high bleeding risk, failed systemic therapy, or shock likely to cause death before systemic thrombolysis can take effect.

xi. Reduced-dose thrombolytic therapy has been investigated as a strategy to minimize bleeding risk.
   (a) Several trials have evaluated "half-dose" alteplase compared with full dose, showing improved hemodynamic parameters and potentially less bleeding, but all trials had severe methodological limitations that limit generalizability.
   (b) A retrospective database of 3768 patients showed that patients who received half-dose alteplase were more likely to require treatment escalation (53.8% vs. 41.4%; $p<0.01$) driven by secondary thrombolysis without notable difference in bleeding complications.
   (c) Given the available data, experts do not recommend consideration for half-dose in PE management. For patients who need reperfusion treatment because of hemodynamic decompensation but who have absolute or relative contraindications to systemic fibrinolysis, CDT is recommended.
   (d) The PEITHO-3 trial (NCT04430569) aims to compare the efficacy and safety of reduced-dose alteplase with standard heparin in patients with intermediate-high risk PE. Results are expected to have a major impact on risk-adjusted acute PE treatment and guideline recommendations.

xii. Patients with a massive or submassive PE should be considered for surgical embolectomy or catheter thrombectomy if they (1) have an unacceptably high risk of bleeding if administered thrombolytics, (2) remain unstable despite thrombolytic administration, or (3) have shock likely to cause death within hours (before the onset of systemic thrombolytics).

xiii. Unless contraindicated, all patients should also receive a parenteral anticoagulant.
   (a) Intravenous unfractionated heparin is recommended over alternative agents for patients in whom thrombolytic therapy is being considered or planned.
   (b) The alteplase package insert recommends holding heparin during the alteplase infusion and reinstituting it when the aPTT returns to less than 2 times the upper limit of normal.
   (c) However, in some clinical trials, heparin was continued during thrombolytic administration.
   (d) Therefore, concurrent heparin administration with thrombolytic therapy in practice varies, and this decision should be individualized on the basis of risk assessment (e.g., bleeding vs. hemodynamics). If anticoagulation is contraindicated, an inferior vena cava filter should be placed.

D. A study highlighted the racial and socioeconomic disparities in the management and outcomes of acute PE in the United States. This study analyzed data from the Nationwide Inpatient Sample for 2016–2018 and identified 1,124,204 hospitalizations for acute PE among U.S. adults, with the disease of 66,570 patients (5.9%) classified as high risk. The study found that the hospitalization rate for PE increased across racial and ethnic groups, with Black people having the highest rate at 20.1 per 10,000 person-years, followed by White people at 13.1 per 10,000 person-years.
   1. The authors reported that Black patients were 13% less likely to receive advanced PE therapy (thrombolysis, catheter-directed treatment, surgical embolectomy, or extracorporeal membrane oxygenation) than White patients (OR 0.87; 95% CI, 0.82–0.93; $p<0.001$), and Asian patients were 24% less likely than others (OR 0.76; 95% CI, 0.70–0.83; $p<0.001$). Patients with Medicare insurance were 27% less likely (OR 0.73; 95% CI, 0.70–0.76; $p<0.001$), Patients with Medicaid insurance were 32% less likely (OR 0.68; 95% CI, 0.64–0.71; $p<0.001$), and patients with "other-insurance" (including patients without insurance) were 14% less likely (OR 0.86; 95% CI 0.81–0.91; $p<0.001$) to receive advanced therapy than patients with private insurance.

2. With respect to mortality, Hispanic patients had a 10% higher mortality rate than White patients (OR 1.10; 95% CI, 1.03–1.18; p=0.005), and Asian patients had a 50% higher mortality rate than others (OR 1.50; 95% CI, 1.35–1.65; p<0.001). Patients with the lowest income also had a 9% greater mortality rate (OR 1.09; 95% CI, 1.04–1.14; p=0.001). Among patients with high-risk PE, Black, Hispanic, and Asian patients had an 11%–50% higher mortality rate than White patients, independent of other factors (p<0.05 for all racial/ethnic groups).

Patient Cases

7. A 48-year-old woman (weight 75 kg) presents to the ED with shortness of breath. The patient's hypoxemia does not improve with supplemental oxygen, and her chest radiograph is not significant for any lung abnormalities. A contrasted chest CT scan reveals a PE in the subsegmental branch of the right pulmonary artery and no RV dilation. The patient's vital signs and significant laboratory values are as follows: heart rate 118 beats/minute, blood pressure 98/62 mm Hg, urinary output 1 mL/kg/hour, troponin T 0.06 ng/mL, brain natriuretic peptide 60 pg/mL, lactate 0.9 mmol/L, and SCr 1.1 mg/dL. In addition to initiating a parenteral anticoagulant, which is best for the patient?
   A. Tenecteplase 40-mg bolus.
   B. Alteplase 100-mg infusion over 2 hours.
   C. Alteplase 50-mg bolus.
   D. No thrombolytic therapy.

8. A 56-year-old man (weight 140 kg) with a history of smoking and chronic obstructive pulmonary disease was admitted to the medical ICU with sudden-onset dyspnea, chest pain, and hypoxemia (Pa_{O_2} 78%). A chest CTA reveals a subsegmental PE. His blood pressure is 87/56 mm Hg, now requiring norepinephrine. Which is the next best step to evaluate and/or treat his PE?
   A. Administer tenecteplase 40-mg intravenous push, followed by a therapeutic heparin infusion.
   B. Obtain an echocardiogram and cardiac enzymes to guide further risk stratification.
   C. Administer catheter-directed alteplase 1 mg/hour for 12 hours by EKOS EndoWave system catheter.
   D. Administer alteplase 100 mg infused over 2 hours, followed by a therapeutic heparin infusion.

REFERENCES

Hypovolemic Shock

1. Alvarado R, Chung KK, Cancio LC, et al. Burn resuscitation. Burns 2009;35:4-14.
2. Baharoglu MI, Cordonnier C, Salman RA, et al. Platelet transfusion versus standard of care after acute stroke due to spontaneous cerebral hemorrhage associated with antiplatelet therapy. Lancet 2016;387:2605-13.
3. Boffard KD, Riou B, Warren B, et al. Recombinant factor VIIa as adjunctive therapy for bleeding control in severely injured trauma patients: two parallel randomized, placebo-controlled, double-blind clinical trials. J Trauma 2005;59:8-18.
4. Bourke DL, Smith TC. Estimating allowable hemodilution. Anesthesiology 1974;41:609-12.
5. Bouzat P, Charbit J, Abback PS, et al. Efficacy and safety of early administration of 4-factor prothrombin complex concentrate in patients with trauma at risk of massive transfusion: the PROCOAG randomized clinical trial. JAMA 2023;329:1367-75.
6. Brohi K, Singh J, Heron M, et al. Acute traumatic coagulopathy. J Trauma Injury Infect Crit Care 2003;54:1127-30.
7. Cannon JW. Hemorrhagic shock. N Engl J Med 2018;378:370-9.
8. Cannon JW, Khan MA, Raja AS, et al. Damage control resuscitation in patients with severe traumatic hemorrhage: a practice management guideline from the Eastern Association for the Surgery of Trauma. J Trauma Acute Care Surg 2017;82:605-17.
9. Carney N, Totten AM, O'Reilly C, et al. Guidelines for the management of severe traumatic brain injury, fourth edition. Neurosurgery 2017;80:164-71.
10. Committee on Trauma: Advanced Trauma Life Support Manual. Chicago: American College of Surgeons, 1997:103-12.
11. Connolly SJ, Crowther M, Eikelboom JW, et al. Full study report of andexanet alfa for bleeding associated with factor Xa inhibitors. N Engl J Med 2019;380:1326-35.
12. CRASH-2 Trial Collaborators. Effects of tranexamic acid on death, vascular occlusive events, and blood transfusion in trauma patients with significant haemorrhage (CRASH-2): a randomised, placebo-controlled trial. Lancet 2010;376:23-32.
13. CRASH-3 Trial Collaborators. Effects of tranexamic acid on death, disability, vascular occlusive events and other morbidities in patients with acute traumatic brain injury (CRASH-3): a randomised, placebo-controlled trial. Lancet 2019;394:1713-23.
14. Dager WE. Developing management plan for oral anticoagulation reversal. Am J Health Syst Pharm 2013;70:S21-31.
15. De Backer D, Biston P, Devriendt J, et al. Comparison of dopamine and norepinephrine in the treatment of shock. N Engl J Med 2010;362:779-89.
16. Delano MJ, Rizoli SB, Rhind SG, et al. Prehospital resuscitation of traumatic hemorrhagic shock with hypertonic solutions worsens hypocoagulation and hyperfibrinolysis. Shock 2015;44:25-31.
17. Eriksson BI, Quinlan DJ, Weitz JI. Comparative pharmacodynamics and pharmacokinetics of oral direct thrombin and factor Xa inhibitors in development. Clin Pharmacokinet 2009;48:1-22.
18. Franchini M, Lippi G. Prothrombin complex concentrates: an update. Blood Transfus 2010;8:149-54.
19. Fröhlich M, Mutschler M, Caspers M, et al. Trauma-induced coagulopathy upon emergency room arrival: still a significant problem despite increased awareness and management? Eur J Trauma Emerg Surg 2019;45:115-24.
20. Frontera JA, Lewin JJ, Rabinstein AA, et al. Guidelines for reversal of antithrombotics in intracranial hemorrhage. Neurocrit Care 2016;24:6-46.
21. Garcia-Tsao G, Abraldes JG, Berzigotti A, et al. Portal hypertensive bleeding in cirrhosis: risk stratification, diagnosis, and management: 2016 practice guidelines by the American Association for the Study of Liver Disease. Hepatology 2017;65:310-35.
22. Gayet-Ageron A, Prieto-Merino D, Ker K, et al. Effect of treatment delay on the effectiveness and safety of antifibrinolytics in acute severe hemorrhage: a meta-analysis of individual patient level data from 40,138 bleeding patients. Lancet 2018;391:125-32.
23. Godier A, Samama CM, Susen S. Plasma/platelets/red blood cell ratio in the management of the bleeding traumatized patient: does it matter? Curr Opin Anesth 2012;25:242-7.

24. Goldstein JN, Refaai MA, Milling TJ, et al. 4PCC vs. plasma for rapid vitamin k antagonist reversal in patients needed urgent surgery or invasive interventions. Lancet 2015;385:2077-87.
25. Gonzalez E, Moore EE, Moore H, et al. Goal directed resuscitation of trauma induced coagulopathy: a pragmatic randomized clinical trial comparing a viscoelastic assay to conventional coagulation assay. Ann Surg 2016;263;1051-9.
26. Gruen RL, Brohi LK, Schreiber M, et al. Haemorrhage control in severely injured patients. Lancet 2012;380:1099-108.
27. Gutierrez G, Reines HD, Wulf-Gutierrez ME. Clinical review: hemorrhagic shock. Crit Care 2004;8:373-81.
28. Haider AH, Chang DC, Efron DT, et al. Race and insurance status as risk factors for trauma mortality. Arch Surg 2008;143:945-9.
29. Harrois A, Hamada SR, Duranteau J. Fluid resuscitation and vasopressors in severe trauma patients. Curr Opin Crit Care 2015;20:632-7.
30. Hauser CJ, Boffard K, Dutton R, et al. Results of the CONTROL trial: efficacy and safety of recombinant activated factor VII in the management of refractory traumatic hemorrhage. J Trauma 2010;69:489-500.
31. Holcomb JB, Tilley BC, Baraniuk S, et al. Transfusion of plasma, platelets, and red blood cells in a 1:1:1 vs. a 1:1:2 ratio and mortality in patients with severe trauma: the PROPPR randomized clinical trial. JAMA 2015;313:471-82.
32. Innerhofer P, Fries D, Mittermayr M, et al. Reversal of trauma-induced coagulopathy using first-line coagulation factor concentrates of fresh frozen plasma (RETIC): a single-centre, parallel-group, open-label, randomized trial. Lancet Haematol 2017;4:258-71.
33. Kaatz S, Kouides PA, Garcia DA, et al. Guidance on the emergent reversal of oral thrombin and factor Xa inhibitors. Am J Hematol 2012;87(suppl 1):S141-5.
34. Kauvar DS, Lefering R, Wade CE. Impact of hemorrhage on trauma outcome: an overview of epidemiology, clinical presentations, and therapeutic considerations. J Trauma 2006;60:S3-11.
35. Kcentra (prothrombin complex concentrate, human) [prescribing information]. CSL Behring, April 2013.
36. Lacroix J, Hebert PC, Fergusson DA, et al. Age of transfused blood in critically ill patients. N Engl J Med 2015;372:1410-8.
37. Madsen TE, Simmons J, Choo EK, et al. The DISPARITY study: do gender differences exist in Surviving Sepsis Campaign resuscitation bundle completion, completion of individual bundle elements, or sepsis mortality? J Crit Care 2014;29:473.e7-11.
38. McKinley BA, Kozar RA, Cocacnour CS, et al. Normal versus supranormal oxygen delivery goals in shock resuscitation: the response is the dame. J Trauma 2002;53:825-32.
39. Meizoso JP, Dudaryk R, Mulder MB, et al. Increased risk of fibrinolysis shutdown among severely injured patients receiving tranexamic acid. J Trauma Acute Care Surg 2018;84:426-32.
40. Milling TJ, Refaai MA, Sarode R, et al. Safety a four-factor prothrombin complex concentrate versus plasma for vitamin K antagonist reversal. Acad Emerg Med 2016;23:466-75.
41. Moore HB, Moore EE, Chapman MP, et al. Plasma first resuscitation to treat hemorrhagic shock during emergency ground transportation in an urban area: a randomized trial. Lancet 2018;392:281-91.
42. Morrison JJ, Ross JD, Dubose JJ, et al. Association of cryoprecipitate and tranexamic acid with improved survival following wartime injury: findings from the MATTERs II study. JAMA Surg 2013;148:218-25.
43. Murphy GJ, Pike K, Rogers CA, et al. Liberal or restrictive transfusion after cardiac surgery. N Engl J Med 2015;372:997-1008.
44. Myles PS, Smith JA, Forbes A, et al. Tranexamic acid in patients undergoing coronary-artery surgery. N Engl J Med 2016;376:136-48.
45. Neal MD, Hoffman MK, Cuschieri J, et al. Crystalloid to packed red blood cell transfusion ratio in the massively transfused patient: when a little goes a long way. J Trauma Acute Care Surg 2012;72:892-8.
46. Nutescu EA, Dager WE, Kalus JS, et al. Management of bleeding and reversal strategies for oral anticoagulants: clinical practice considerations. Am J Health Syst Pharm 2013;70:e82-97.
47. O'Shea RS, Davitkov P, Ko CW, et al. AGA clinical practice guideline on the management of coagulation disorders in patients with cirrhosis. Gastroenterology 2021;161:1615-1627.e1.

48. Panteli M, Pountos I, Giannoudis PV. Pharmacologic adjuncts to stop bleeding. Eur J Trauma Emerg Surg 2016;42:303-10.
49. Peltan ID, Vande Vusse LK, Maier RV, et al. An international normalized ratio-based definition of acute traumatic coagulopathy is associated with mortality, venous thromboembolism, and multiple organ failure after injury. Crit Care Med 2015;43:1429-38.
50. Pham TN, Cancio LC, Gibran NS. American Burn Association practice guidelines burn shock resuscitation. J Burn Care Res 2008;29:259-66.
51. Pollack CV, Reilly PA, Ryn J, et al. Idarucizumab for dabigatran reversal – full cohort analysis. N Engl J Med 2017;377:431-41.
52. Porter JM, Ivatury RR. In search of the optimal end points of resuscitation in trauma patients: a review. J Trauma 1998;44:908-14.
53. Prochaska M, Salcedo J, Berry G, et al. Racial differences in red blood cell transfusion in hospitalized patients with anemia. Transfusion 2022;62:1519-26.
54. Sarode R, Milling TJ, Refaai MA, et al. Efficacy and safety of 4PCC in patients on vitamin K antagonists presenting with major hemorrhage. Circulation 2013;128:1234-43.
55. Siegal DM, Curnutte JT, Connolly SJ, et al. Andexanet alfa for the reversal of factor Xa inhibitor activity. N Engl J Med 2015;373:2413-24.
56. Sihler KC, Napolitano LM. Complications of massive transfusion. Chest 2010;137:209-20.
57. Sims CA, Holena D, Kim P, et al. Effect of low-dose supplementation of arginine vasopressin on need for blood product transfusions in patients with trauma and hemorrhagic shock. A randomized clinical trial. *JAMA Surg* 2019;154:994-1003.
58. Spahn DR, Bouillon B, Cerny V, et al. The European guideline on management of major bleeding and coagulopathy following trauma: fifth edition. Crit Care 2019;23:98.
59. Sperry JL, Guyette FX, Brown JB, et al. Prehospital plasma during air medical transport in trauma patients at risk for hemorrhagic shock. N Engl J Med 2018;26:315-26.
60. Tomaselli GF, Mahaffey KW, Cuker A, et al. 2020 ACC expert consensus decision pathway on management of bleeding in patients on oral anticoagulants: a report of the American College of Cardiology Solution Set Oversight Committee. J Am Coll Cardiol 2020;76:594-622.
61. Tripodi A, Mannucci PM. The coagulopathy of chronic liver disease. N Engl J Med 2011;365:147-56.
62. Villaneuva C, Colomo A, Bosch A, et al. Transfusion strategies for acute gastrointestinal bleeding. N Engl J Med 2013;368:11-21.
63. WOMAN Trial Collaborators. Effect of early tranexamic acid administration on mortality, hysterectomy, and other morbidities in women with post-partum haemorrhage (WOMAN): an international, randomized, double-blind, placebo-controlled trial. Lancet 2017;389;2105-16.
64. Yank V, Tuohy CV, Logan AC, et al. Systematic review: benefits and harms of in-hospital use of recombinant factor VIIa for off-label indications. Ann Intern Med 2011;154:529-40.

Obstructive Shock

1. Angle MR, Molloy DW, Penner B, et al. The cardiopulmonary and renal hemodynamic effects of norepinephrine in canine pulmonary embolism. Chest 1989;95:1333-7.
2. Berg KM, Soar J, Andersen LW, et al. Adult advanced life support: 2020 international consensus on cardiopulmonary resuscitation and emergency cardiovascular care science with treatment recommendations. Circulation 2020;142:S92-S139.
3. Chatterjee S, Weinberg I, Yeh RW, et al. Risk factors for intracranial haemorrhage in patients with pulmonary embolism treated with thrombolytic therapy. Development of the PE-CH score. Thromb Haemost 2017;117:246-51.
4. Curtis GC, Lam SW, Reddy AJ, Bauer SR. Risk factors associated with bleeding after alteplase administration for pulmonary embolism: a case-control study. Pharmacotherapy 2014;34:818-25.
5. Daley MJ, Lat I. Clinical controversies in thrombolytic therapy for the management of acute pulmonary embolism. Pharmacotherapy 2012;32:158-72.
6. Farmakis I, Cushman M, Valerio L, et al. Social determinants of health and pulmonary embolism treatment and mortality: the Nationwide Inpatient Sample. Presented at: 64th American Society of Hematology Annual Meeting and Exposition; December 10, 2022; New Orleans, LA.
7. Greyson CR. Pathophysiology of right ventricular failure. Crit Care Med 2008;36:S57-65.

8. Hoeper MM, Granton J. Intensive care unit management of patients with severe pulmonary hypertension and right heart failure. Am J Respir Crit Care Med 2011;184:1114-24.
9. Jaff MR, McMurtry MS, Archer SL, et al. Management of massive and submassive pulmonary embolism, iliofemoral deep vein thrombosis, and chronic thromboembolic pulmonary hypertension: a scientific statement from the American Heart Association. Circulation 2011;123:1788-830.
10. Kearon C, Akl EA, Comerota AJ, et al. Antithrombotic therapy for VTE disease: Antithrombotic Therapy and Prevention of Thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. Chest 2012;141:e419S-94S.
11. Kerbaul F, Rondelet B, Motte S, et al. Effects of norepinephrine and dobutamine on pressure load-induced right ventricular failure. Crit Care Med 2004;32:1035-40.
12. Kerber RE, Gascho JA, Litchfield R, et al. Hemodynamic effects of volume expansion and nitroprusside compared with pericardiocentesis in patients with acute cardiac tamponade. N Engl J Med 1982;307:929-31.
13. Kiser TH, Burnham EL, Clark B, et al. Half-dose versus full-dose alteplase for treatment of pulmonary embolism. Crit Care Med 2018;46:1617-25.
14. Konstantinides S. Clinical practice. Acute pulmonary embolism. N Engl J Med 2008;359:2804-13.
15. Konstantinides SV, Barco S, Lankeit M, et al. Management of pulmonary embolism: an update. J Am Coll Cardiol 2016;67:976-90.
16. Konstantinides SV, Meyer G, Becattini C, et al. 2019 ESC guidelines for the diagnosis and management of acute pulmonary embolism developed in collaboration with the European Respiratory Society (ERS). Eur Heart J. 2020;41:543-603.
17. Konstantinides SV, Vicaut E, Danays T, et al. Impact of thrombolytic therapy on the long term outcome of intermediate risk pulmonary embolism. J Am Coll Cardiol 2017;69:1536-44.
18. Kucher N, Goldhaber SZ. Management of massive pulmonary embolism. Circulation 2005;112:e28-32.
19. Maisch B, Seferovic PM, Ristic AD, et al. Guidelines on the diagnosis and management of pericardial diseases executive summary; the task force on the diagnosis and management of pericardial diseases of the European Society of Cardiology. Eur Heart J 2004;25:587-610.
20. Meyer G, Vicaut E, Danays T, et al. Fibrinolysis for patients with intermediate-risk pulmonary embolism. N Engl J Med 2014;370:1402-11.
21. Mostafa A, Briasoulis A, Telila T, et al. Treatment of massive or submassive acute pulmonary embolism with catheter directed thrombolysis. Am J Cardiol 2016;117:1014-20.
22. Spöhr F, Arntz HR, Bluhmki E, et al. International multicentre trial protocol to assess the efficacy and safety of tenecteplase during cardiopulmonary resuscitation in patients with out-of-hospital cardiac arrest: the Thrombolysis in Cardiac Arrest (TROICA) study. Eur J Clin Invest 2005;35:315-23.
23. Stein PD, Matta F. Thrombolytic therapy in unstable patients with acute pulmonary embolism: saves lives but underused. Am J Med 2012;125:465-70.
24. Stevens SM, Woller SC, Kreuziger LB, et al. Antithrombotic therapy for VTE disease: second update of the CHEST Guideline and Expert Panel Report. Chest 2021;160:e545-e608.
25. Todd JL, Tapson VF. Thrombolytic therapy for acute pulmonary embolism: a critical appraisal. Chest 2009;135:1321-9.
26. Wan S, Quinlan DJ, Agnelli G, et al. Thrombolysis compared with heparin for the initial treatment of pulmonary embolism: a meta-analysis of the randomized controlled trials. Circulation 2004;110:744-9.

ANSWERS AND EXPLANATIONS TO PATIENT CASES

1. **Answer: D**

This patient has class II hemorrhagic shock caused by penetrating trauma (heart rate greater than 100 beats/minute, respiratory rate 20–30 breaths/minute, and anxious on examination). The appropriate resuscitation strategy should focus on selecting the fluid, volume, and resuscitation goal. Recommended end points for resuscitation include SBP greater than 90 mm Hg, urinary output greater than 30 mL/hour, and normal mentation. Administering fluid to target an SBP greater than 90 mm Hg would be appropriate (Answer A is incorrect). Blood products for transfusion are indicated when the patient's estimated blood loss is greater than 30% (Answers B and C are incorrect because the patient has class II hemorrhagic shock). Given his hemorrhagic shock class, the patient may be initially managed with crystalloid boluses targeting a urinary output greater than 30 mL/hour, an SBP greater than 90 mm Hg, and normal mentation (Answer D is correct).

2. **Answer: B**

Using the Parkland formula for burn resuscitation, the 24-hour fluid requirement would be 13.6 L, to be segmented into the first 8 hours of resuscitation and the remaining 16 hours. Calculating fluid requirements according to weight and TBSA to target a urinary output of greater than 0.5 mL/kg/hour would be correct (Answers A, C, and D are incorrect). Answer B is correct because the total volume requirement accounting for prehospital fluid, titration of fluid rates, and target urinary output are appropriate.

3. **Answer: D**

Given this patient's hemodynamic parameters in a positive FAST examination, he will likely require a massive transfusion, as assessed by the ABC (Assessments of Blood Consumption) score for a massive transfusion protocol, which should be initiated promptly. A fixed ratio of 1:1:1 PRBC, plasma, and platelets is more likely to achieve hemostasis and lower mortality from exsanguination at 24 hours than is 2:1:1 (Answer D is correct). Although fluid resuscitation is an adjunct to increase intravascular volume and tissue perfusion, 0.9% sodium chloride does not increase oxygen-carrying capacity, and total volume should be minimized to less than 1.5 L (Answer A is incorrect). In a traumatic hemorrhage, a normal hemoglobin likely will not reflect the extent of blood loss secondary to hemoconcentration until capillary recruitment is complete or resuscitation begins. In addition, resuscitation efforts in a hemodynamically unstable patients should not be delayed while awaiting laboratory results (Answer B is incorrect). Although vasopressors may be considered an adjunct to maintain tissue perfusion while minimizing fluid volume, this strategy should be reserved for those with life-threatening hypotension despite resuscitation efforts (Answer C is incorrect).

4. **Answer: B**

Acute hypocalcaemia is a common complication in a massive transfusion secondary to citrate added to stored blood. Calcium is essential for stabilization of fibrin, and low ionized calcium concentrations in traumatic hemorrhage are associated with increased mortality. Therefore, ionized calcium concentrations should be maintained within the normal range (Answer B is correct). Evidence is insufficient to recommend sodium bicarbonate therapy in patients with acute trauma and has the potential for increased mortality. The patient's acidosis should be corrected by addressing the underlying cause – in this case, hemorrhage (Answer A is incorrect). Use of rFVIIa should only be considered if major bleeding and persistent traumatic coagulopathy persist despite best practice, inconsistent with the patient's current condition. In addition, a pH less than 7.2 is a predictor of a poor response to rFVIIa (Answer C is incorrect). Unfortunately, this patient's injury was 5 hours ago, and he has no laboratory evidence of hyperfibrinolysis. In this setting, tranexamic acid does not improve mortality after 3 hours from injury and has been associated with an increased risk of death from bleeding (Answer D is incorrect).

5. **Answer: C**

At this point, ongoing resuscitation should be goal directed. The TEG reveals hypofibrinolysis or "shutdown," not hyperfibrinolysis for which tranexamic acid is expected to help. In addition, if there was no TEG to evaluate for hypofibrinolysis, tranexamic acid should be administered within the first 3 hours from injury, and this patient's injuries occurred more than 4 hours earlier (Answers B and D are incorrect). With the updated history that the patient was taking dabigatran preinjury, dabigatran is likely contributing to the patient's ongoing

coagulopathy and hemorrhage. A normal TEG and elevated aPTT may not rule out the presence of dabigatran. In addition, it seems the patient took the last dabigatran dose within the previous 12 hours. Therefore, dabigatran reversal is indicated. Idarucizumab is FDA approved for dabigatran reversal and recommended in various guidelines, whereas PCCs have weak supporting evidence and are recommended only if idarucizumab is unavailable (Answer C is correct; Answer A is incorrect).

6. **Answer: B**

This patient has an acute GI hemorrhage complicated by supratherapeutic INR warfarin and hemodynamic instability necessitating warfarin reversal with 4F-PCC. Given his weight and degree of INR elevation, the package insert dose is 50 units/kg not to exceed 5000 units (Answer B is correct; Answer D is incorrect). Although fixed, low-dose PCC has been evaluated for warfarin reversal, this dose is off-label and supported by a lower quality of evidence. In addition, the 1000-unit dose has shown lower achievement of a goal INR, necessitating rescue doses, and a low dose would probably not achieve adequate laboratory reversal, given his weight and excessively high INR (Answer A is incorrect). Finally, although rFVIIa lowers INR, it is no longer recommended for the reversal of vitamin K antagonists, given the incomplete correction of factors II, IX, and X and the possibility that the INR does not reflect the patient's underlying coagulopathy (Answer C is incorrect).

7. **Answer: D**

The patient has evidence of a PE, but she lacks features of an increased risk of early mortality from it. She does not have shock or evidence of end-organ hypoperfusion and thus does not have a massive PE. In addition, she has no evidence of RV dysfunction (no RV dilation on chest CT, brain natriuretic peptide less than 90 pg/mL) or myocardial necrosis (troponin less than 0.1 mg/mL) and thus does not have a submassive PE. The patient is best classified as having a low-risk PE. A meta-analysis suggested that thrombolytics do not decrease mortality in unselected (and low risk) patients and may increase bleeding risk (Answer D is correct; Answers A–C are incorrect).

8. **Answer: D**

The patient has a massive or high-risk PE, as evidenced by signs of hemodynamic complications requiring norepinephrine, likely confounded by poor physiologic reserve with chronic obstructive pulmonary disease. The current guidelines recommend systemic thrombolysis in patients with a massive PE and an acceptable risk of bleeding. This patient does not appear to have any obvious risk of bleeding, including age; therefore, systemic thrombolytic agents are indicated in addition to a therapeutic heparin infusion. The most common options include alteplase as a fixed dose of 100 mg infused over 2 hours or tenecteplase adjusted according to patient weight. Because this patient weighs 140 kg, the recommended tenecteplase dose would be a 50-mg intravenous push once (Answer A is incorrect); therefore, alteplase 100 mg infused over 2 hours is the most appropriate answer (Answer D is correct). The 2021 CHEST guidelines state that if thrombolytic agents are indicated, systemic thrombolysis is preferred to catheter-directed thrombolytic administration (Answer C is incorrect). Echocardiogram and cardiac enzymes are not considered necessary to guide therapy in patients with hypotension/shock and a CTA-confirmed PE (Answer B is incorrect).

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: C**

This patient presents with likely hemorrhagic shock after blunt trauma. The extent of injuries in highly vascularized areas suggests a hemorrhagic source. Given the extent of confusion, oliguria, tachycardia, and tachypnea, as well as the amount of blood loss (35% of total blood volume), this patient has class III hemorrhage (Answer C is correct; Answers A, B, and D are incorrect).

2. **Answer: C**

The Fick equation describes oxygen delivery (Do_2 (mL/minute) = 10 x CO (L/minute) x Cao_2). A diagnosis of shock is typically based on hemodynamic, clinical, and biochemical assessment of impaired Do_2. Although an elevated INR may represent a coagulopathy that can contribute to bleeding, it is not directly related to impaired Do_2. In addition, hematemesis is a clinical sign of bleeding but is not necessarily specific to a shock syndrome (Answer A is incorrect). Although heart rate is a determinant of CO, CO is not generally impaired unless the patient has acute supraventricular or ventricular tachycardia. In addition, an elevated serum creatinine may be a biochemical sign of impaired tissue oxygenation, but this generally presents in a delayed fashion and may represent chronic renal dysfunction in this specific patient (Answer B is incorrect). A low $Scvo_2$ represents the balance between oxygen consumption and delivery (not solely delivery) and does not explain reduced Do_2. Although an elevated lactate concentration may indicate impaired Do_2 in shock syndrome, lactate may be elevated secondary to altered hepatic metabolism and is not a specific marker for circulatory shock (Answer D is incorrect). Therefore, according to the Fick equation, a low hemoglobin reduces the Cao_2, decreasing Do_2. Cold and clammy extremities is a common clinical marker of a shock syndrome, indicating a low-flow state (Answer C is correct).

3. **Answer: D**

This patient presents with likely hemorrhagic shock after blunt trauma. The patient has signs consistent with intra-abdominal bleeding with the positive FAST examination in hemodynamic instability; thus, urgent intervention is required. According to the extent of altered mentation, tachycardia, and tachypnea, this patient has class IV hemorrhage, consistent with greater than 40% blood loss, indicating the requirement for erythrocyte transfusions. However, giving only erythrocyte transfusions would increase oxygen-carrying capacity but not correct the likely underlying coagulopathy (Answer A is incorrect). Although warmed lactated Ringer solution may be indicated in a trauma patient with bleeding, excessive fluid resuscitation (above 1.5 L) should initially be avoided (Answer B is incorrect). Given the patient's blood pressure, heart rate, and positive FAST, he is at high risk of requiring massive transfusion, which should be initiated promptly with an empiric ratio. In addition, given the presence of hemorrhagic shock requiring erythrocyte transfusions, tranexamic acid is indicated early to improve mortality (Answer D is correct). Although goal-directed resuscitation is recommended for continued resuscitation, empiric blood product transfusions should not be delayed while awaiting laboratory results unless point-of-care assays are available (Answer C is incorrect).

4. **Answer: B**

This patient has apparent warfarin toxicity contributing to an acute GI hemorrhage. Because he meets the criteria for a major bleed, particularly associated with hemodynamic instability, many guidelines recommend anticoagulation reversal. Although phytonadione is the specific reversal for warfarin to promote hepatic production of factors II, VII, IX, and X, it does not immediately correct coagulopathy, and factor replacement is recommended (Answer D is incorrect). In addition, when used for life-threatening hemorrhage, intravenous administration is recommended because of its quicker onset of action of 4–6 hours compared with 18–24 hours for oral administration (Answer A is incorrect). Although plasma has traditionally been used for immediate factor replacement for warfarin reversal, the time to laboratory reversal is inferior to PCC. In addition, the volume of plasma needed to reverse his INR would likely not be well tolerated by this patient with CTEPH (Answer C is incorrect). Therefore, 4F-PCC, which contains nonactivated highly concentrated factors II, VII, IX, and X, is recommended for warfarin reversal in life-threatening hemorrhage because of its superiority for laboratory reversal and noninferiority for clinical hemostasis. Rates of thrombotic events appear similar when comparing 4F-PCC with plasma for warfarin reversal in life-threatening hemorrhage and surgical populations; however, rates

of fluid overload complications are greater with plasma (Answer B is correct).

5. **Answer: A**
The patient has a massive PE, as evidenced by pulseless activity. Massive PE should be treated with thrombolytic therapy unless absolute contraindications to thrombolytics are present. Prolonged chest compressions may be considered a relative contraindication to thrombolytics, but this patient had a short duration of chest compressions (Answer A is correct). Troponin T and brain natriuretic concentrations may help classify a patient as having a submassive PE, but the patient in this case has already fulfilled the criteria for a massive PE, and these laboratory values will not change the patient's treatment (Answers B and C are incorrect). Although a TTE might provide information on RV function, it is unlikely to change the patient's treatment plan with thrombolytics and might delay therapy (Answer D is incorrect).

6. **Answer: B**
This patient's PE is causing stress on the RV, as evidenced by his echocardiogram and positive cardiac enzymes. Therefore, this is not a low-risk PE (Answer D is incorrect). However, the patient is hemodynamically stable at this point, so it is also not a high-risk PE (Answer A is incorrect). Given that the patient has both evidence for RV straining and myocardial ischemia as represented by both the echocardiogram and the cardiac enzymes, this meets the criteria for an intermediate-high risk PE (Answer B is correct; Answer C is incorrect).

7. **Answer: B**
This patient appears to be clinically worsening with blood pressures that are approaching hemodynamic instability and worsening oxygenation requiring high levels of F_{IO_2}. Consideration for reperfusion therapy is warranted. Given his history of a recent GI bleed requiring ICU admission, systemic, full-dose thrombolytics would be contraindicated (Answer A is incorrect). A thrombolytic strategy to minimize the risk of bleeding may be appropriate. Although half-dose alteplase has been studied, it is not recommended because of insufficient evidence (Answer C is incorrect). Therefore, in intermediate-high risk PE with clinical deterioration, the guidelines recommend CDT when systemic thrombolytic contraindications exist and resources are available, such as in an urban, academic ICU. Catheter-directed thrombolysis is administered with mechanical methods (e.g., ultrasound-assisted thrombolysis) and specific catheters (not pulmonary artery catheters) (Answer B is correct; Answer D is incorrect).

8. **Answer: A**
This patient now has a cardiac arrest secondary to his PE. Because he has a confirmed PE as the precipitant of cardiac arrest, systemic thrombolytic is recommended as an emergency treatment. Given the mortality rates from PE in a cardiac arrest, standard contraindications to thrombolysis may be suspended in favor of a lifesaving intervention. Systemic thrombolytics in this setting may be associated with return of spontaneous circulation and possibly survival benefits (Answer D is incorrect). The recommended alteplase dose is a 50-mg intravenous bolus, repeated in 15 minutes if needed (Answer A is correct; Answer C is incorrect). Although surgical embolectomy would be reasonable in the operating room, systemic thrombolytics would be preferred, given the patient's current location in the ICU (Answer B is incorrect).

9. **Answer: D**
This patient is experiencing a high-risk or massive PE, as evidenced by a confirmed CT scan, clinical signs consistent with a PE, and a dilated right ventricle on TTE in the setting of sustained hypotension requiring norepinephrine. Although an inferior vena cava filter may prevent additional embolization, it will not limit the growth or cause lysis of the current clot burden associated with his PE (Answer A is incorrect). For a high-risk or massive PE, administration of systemic thrombolytic agents is recommended unless contraindications are present. Because this patient had a recent subarachnoid hemorrhage, the risk of systemic thrombolytic therapy might outweigh the potential benefit (Answer B is incorrect), even at a reduced dose (Answer C is incorrect). Therefore, the CHEST guidelines recommend CDT for patients who are hemodynamically unstable from a PE but have a high bleeding risk (Answer D is correct).